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Strain-coupled one-dimensional turbulence: a single-line closure for rapid distortion, its measured limits, and the consequence for leading-edge noise

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The rapid distortion of turbulence by a mean strain is central to flows from wind-tunnel contractions to the stagnation regions of lifting surfaces, yet its economical prediction founders on a closure problem: the rapid pressure–strain term is a nonlocal three-dimensional spectral integral, while affordable models evolve at most one-dimensional or single-point information. We present a strain-coupled formulation of one-dimensional turbulence (ODT) and, as its analytical core, a systematic treatment of what a single line of turbulence data determines about the rapid pressure–strain term: the obstruction to a single-line closure is stated exactly, sharp bounds on the rapid term are derived from line data alone, and the closure assumption of axisymmetry about the line is not assumed but measured—it is exact for axisymmetric distortion and its plane-strain error is quantified as a function of accumulated strain. A strained-box direct numerical simulation with an exact rapid-distortion companion provides the independent spectral benchmark: it shows that the correct linear reference for line-projected spectra is broadband rather than a rigid spectral translation, and a three-way comparison localises the model's remaining deficiency in its wavenumber-uniform rapid kinematics, quantified by a distance-from-linear-theory envelope as a function of strain rapidity. Hierarchical application of the model's energy-redistributing kernel to eddy sub-scales supplies the measured missing ingredient—scale-local relaxation—where it reaches the scales of the imbalance, and decoupling the relaxation from the eddy clock removes its deep-strain fade while a size cap preserves the one-point physics: in a five-way comparison against the DNS and exact linear theory on the benchmark observable, the mechanism moves the model's spectral allocation of anisotropy towards the DNS at moderate strain rapidity, with its rapid-strain saturation measured and stated. The output is then carried to the motivating application: supplied to Amiet's leading-edge-noise theory in place of the frozen von Kármán inflow of standard practice, the computed distorted spectrum lowers the predicted radiated level by 5–20 dB at the energy-containing frequencies for total strains 0.5–2, in a geometry-free kernel ratio and in an assembled absolute prediction alike—the magnitude and sign of the

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term that frozen-spectrum practice omits, opposite to the surface-blocking correction with which it must be combined.

Key words:

1. Introduction

The distortion of turbulence by a mean velocity gradient is among the most basic problems of turbulence dynamics. When a turbulent flow is strained—accelerated through a contraction, deflected around a bluff body, compressed towards a stagnation region—the mean deformation reorganises the fluctuating field: it amplifies some velocity components and suppresses others, redistributes energy among them through the fluctuating pressure, and rescales the spectral content of the motion. When the strain accumulates in a time short compared with the eddy turnover time, the nonlinear interactions cannot keep pace and the response is governed, to leading order, by the linear dynamics of rapid distortion, first analysed by Batchelor & Proudman (1954) and Townsend (1976), developed for flow around obstacles by Hunt (1973) and Hunt & Graham (1978), and given its modern systematic form by Hunt & Carruthers (1990) and Sagaut & Cambon (2018). The turbulence that emerges from a strained region differs from the turbulence that entered it—in intensity, anisotropy and spectral shape—and that difference is often the quantity of practical interest, from grid turbulence in wind-tunnel contractions (Mills & Corrsin 1959; Tucker & Reynolds 1968; Lee & Reynolds 1985) to the distortion of oncoming turbulence approaching a lifting surface (Hunt & Graham 1978; Goldstein & Atassi 1976).

Capturing this distortion at a cost compatible with parametric study is difficult. Scale-resolving simulation represents it directly but precludes broad sweeps over intensity, length-scale ratio and Reynolds number; linear rapid-distortion theory (RDT) is exact in the rapid limit but neglects, by construction, the inter-scale transfer and the slow return to isotropy that act over any finite strain duration (Sagaut & Cambon 2018); second-moment closures restore the nonlinear relaxation but discard the spectral information that the distortion redistributes (Launder *et al.* 1975; Pope 2000). Between these sits a gap: a reduced-order model that *evolves* strained turbulence with full scale resolution, retaining the rapid distortion and the nonlinear relaxation together, at negligible cost.

A concrete and demanding instance, and the application motivating this work, is turbulence–aerofoil interaction noise. The prevailing prediction framework (Amiet 1975; Paterson & Amiet 1977; Roger & Moreau 2010) obtains the radiated spectrum as a linear functional of the upwash velocity spectrum of the incoming turbulence—but the spectrum it consumes is prescribed, not computed, usually as an isotropic von Kármán form (von Kármán 1948), whereas the turbulence actually reaching the leading edge is strained turbulence. The importance of this pre-impact distortion for the prediction is by now well established: measurements and scale-resolved simulations show the near-edge spectrum systematically altered (dos Santos *et al.* 2023; Ribeiro *et al.* 2023; Piccolo *et al.* 2024), and corrected predictions have been obtained by prescribing RDT-transformed inflow spectra (de Santana *et al.* 2016; dos Santos *et al.* 2023) or by sampling the spectrum close to the edge (Piccolo *et al.* 2024). What these corrections share is that the distorted spectrum is imposed—from linear theory, or from a measurement that a prediction tool does not have. Synthetic-turbulence generators (Kim & Haeri 2015) share the limitation by construction: they reproduce a prescribed spectrum rather than evolving one. A model that *predicts* the distorted spectrum, including the nonlinear relaxation that linear transforms omit, is the missing element—here and wherever distorted turbulence feeds a downstream prediction, from contraction design to the turbulence ingested by rotors. The subject of this paper is that distortion problem; the acoustic stage enters only as the established linear functional that consumes the model's output.

Map-based stochastic one-dimensional models offer the route. The one-dimensional turbulence (ODT) model (Kerstein 1999; Kerstein *et al.* 2001), built on the triplet map of the linear-eddy model (Kerstein 1991), evolves a three-component velocity field along a notional line, resolving the full range of length scales while representing turbulent advection as a stochastic sequence of mapping events; a kernel mechanism redistributes energy among components, emulating pressure scrambling (Wunsch & Kerstein 2001; Kerstein *et al.* 2001). Generalised to variable density (Ashurst & Kerstein 2005), adaptive meshes (Lignell *et al.* 2013; Stephens & Lignell 2021) and cylindrical geometry (Lignell *et al.* 2018), embedded as a subgrid closure (Schmidt *et al.* 2010), and coupled to acoustic analogies for jet noise (Sharma *et al.* 2022; Medina Méndez *et al.* 2023), ODT is an accurate and inexpensive generator of scale-resolved turbulence statistics—exactly what a model of strained turbulence must deliver. But standard ODT cannot represent distortion by a mean strain: its pressure treatment models only the slow, turbulence-driven return to isotropy, not the rapid part driven directly by the mean strain, and the apparent remedy—importing the RDT amplitude equation, whose pressure response is the solenoidal projection in three-dimensional wavenumber space—is unavailable on a line that resolves only the wavenumber along its own coordinate. Reconciling the inherently three-dimensional, non-local pressure physics of rapid distortion with the one-dimensional, local structure of ODT is the central modelling problem addressed here.

The paper makes six contributions, each stated so as to be falsifiable. First, a strain-coupled formulation of ODT (SC-ODT, hereafter) (§ 2): production enters as an exact continuous forcing, the rapid pressure–strain as an energy-conserving redistribution operator built from an explicit spectral closure, and the wavevector kinematics as a dilatation of the domain; the construction preserves the model's structural invariants and reproduces the rapid-distortion evolution of the component energies, under the stated closure, with no adjustable constant (§ 3). Second, the correct linear reference for what the model actually predicts: exact RDT, projected onto a line, is *broadband*—it does not translate the projected spectra rigidly—so the reference against which any line model must be judged differs from the model's own rigid kinematics already at the linear level (§ 4.2). Third, an independent spectral benchmark: a strained-box direct numerical simulation with exact-RDT companion calculations, reduced to the same one-dimensional observables, yielding a three-way comparison that localises the model's remaining deficiency in its wavenumber-uniform rapid kinematics and a distance-from-linear-theory envelope as a function of strain rapidity Sk_t/ε, the strain rate in eddy-turnover units (§ 4.6). Fourth, an answer to the question of principle—what can a single line say about the rapid pressure–strain term, which is exactly a non-local three-dimensional object?—in a form deliberately more modest than a closure claim: the obstruction is stated exactly, sharp bounds on the rapid term are derived from line data alone, and the axisymmetry hypothesis under which the term collapses is not assumed but *measured*, exact for axisymmetric distortion and with a quantified error under plane strain (§ 5). Fifth, the measured limits of the model and the mechanism that moves them (§ 6): a transmission diagnostic shows the strain-induced anisotropy crossing two triplet-map generations—a factor of nine in scale—undiminished, and a controlled family of interventions falsifies acceptance gating and map reweighting as remedies. Hierarchical application of the energy-redistributing kernel to eddy sub-scales—scale-local relaxation—supplies the missing ingredient where it reaches the scales of the imbalance; the mechanism's own deep-strain fade is traced to its event-locked timing and removed by an independent, size-capped relaxation clock, leaving a single measured allocation bound. Sixth, the consequence for the motivating application (§ 7): the computed distorted spectrum is supplied to Amiet's leading-edge-noise theory in place of the frozen inflow of standard practice, both as a ratio from which aerofoil geometry and observer cancel exactly and as an assembled absolute prediction at the reference operating point, quantifying the 5–20 dB term that prescribed-spectrum practice omits; the assembled chain is confronted with four published experiments, which it reproduces to 1–2 dB rms where the facilities are cleanest. Throughout, the posture is the same: where the model is right we show against what, and where it is wrong we measure by how much and locate why.

The paper falls into three parts. The first is the model and what a line can know: § 2 reviews standard ODT and derives the strain-coupled formulation; § 3 verifies it against exact rapid-distortion solutions and validates it at the moment level against the DNS of Lee & Reynolds (1985); § 4 constructs the linear reference, examines the emergent distorted spectrum and its operating point, and carries out the strained-box DNS benchmark; § 5 treats the single-line closure question. The second part, § 6, measures the model's limits and establishes scale-local relaxation—timed by its own clock—as the mechanism that moves them. The third part, § 7, carries the computed distortion to the leading-edge-noise input. Section 8 concludes. Supplementary material contains the validation against the strained-turbulence experiment of Chen *et al.* (2006) and the complete intervention family behind the mechanism ledger.

2. Strain-coupled one-dimensional turbulence

The linear rapid-distortion equations, recalled below, contain two distinct ingredients: an evolution of the Fourier *amplitudes*, comprising production and the solenoidal pressure projection, and an evolution of the *wavevectors* that carries the kinematic straining of scales. The pressure part of the amplitude equation requires the full three-dimensional wavevector and is therefore not directly representable on a one-dimensional line; that incompatibility organises this section. The distortion physics is split according to its structure and each part assigned to the mechanism that can carry it faithfully: the production, which is local in physical space, is imposed as a continuous forcing of the line velocity (§ 2.3); the rapid pressure–strain, which is not, is supplied as a continuous, energy-conserving redistribution operator constructed to reproduce the homogeneous rapid-distortion evolution of the component energies under an explicit spectral closure (§ 2.4); and the wavevector kinematics are recovered by a consistent straining of the domain (§ 2.5). The slow, turbulence-driven pressure–strain and the inertial cascade remain the discrete eddy events of the standard model, reviewed first.

Throughout, $A_{ij} = \partial U_i/\partial x_j$ denotes the imposed mean velocity gradient, taken uniform across the line at each instant; the medium is incompressible, $A_{ii} = 0$. The line carries the fluctuation field $u_i(y,t)$ about this mean; $R_{ij} \equiv \overline{u_i u_j}$ is the Reynolds-stress tensor, the overline denoting the average over the statistically homogeneous directions (equivalently, the ensemble of independent realisations), and $k_t \equiv \frac{1}{2}R_{ii}$ the turbulent kinetic energy. In the homogeneous configurations used to establish and test the formulation the spatial line average and the ensemble average coincide.

2.1. *The standard model*

This subsection reviews the one-dimensional turbulence model of Kerstein (1999) in the vector-valued formulation of Kerstein *et al.* (2001), in the notation used throughout; nothing in it is original. The model evolves a three-component velocity field $u_i(y,t)$, $i = 1,2,3$, on a one-dimensional domain whose coordinate y is identified with the direction x_2. The fields represent individual realisations; ensemble statistics are accumulated over many independent simulations. Two interleaved mechanisms advance the state: between events the fields obey ordinary one-dimensional diffusion, $(\partial_t - \nu\,\partial_y^2)\,u_i = 0$, with kinematic viscosity ν; and a stochastic sequence of instantaneous *eddy events* represents the effect of individual turbulent overturns.

Because a continuum rotational motion cannot be represented on a line, an eddy event is an instantaneous, measure-preserving rearrangement of the profiles together with an additive velocity change,

$$u_i(y) \;\longrightarrow\; u_i\big(f(y)\big) + c_i\,K(y), \tag{2.1}$$

where f is the *triplet map* (Kerstein 1991, 1999): on an interval $[y_0, y_0 + l]$ of size l it compresses the profile to one third of its length, inserts three copies, reverses the central copy, and leaves the exterior unchanged. It is *measure preserving* (the non-local analogue of a divergence-free rearrangement), *continuous*, and *scale local* (it changes gradients by at most an order-unity factor) (Kerstein *et al.* 2001)—the three invariants whose preservation under an imposed mean strain is established in § 2.3. The additive term is built from the *kernel*

$$K(y) \equiv y - f(y), \qquad \int K(y)\,\mathrm{d}y = 0, \qquad \int K^2(y)\,\mathrm{d}y = \tfrac{4}{27}\,l^3, \tag{2.2}$$

non-zero only within the eddy interval; the first identity guarantees that the additive change conserves the momentum of each component, and the second fixes the energetics of the redistribution.

The kernel term is the model surrogate for the pressure-mediated redistribution of energy among velocity components (Wunsch & Kerstein 2001; Kerstein *et al.* 2001). Requiring that an event redistribute kinetic energy among components while conserving the total, with a rule invariant under permutation of the component indices, gives

$$\varDelta E_i = \alpha \sum_j \mathsf{T}_{ij}\,Q_j, \qquad \mathsf{T} = \frac{1}{2}\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \qquad Q_i \equiv \tfrac{27}{8}\,\rho_0\,l\,u_{i,K}^2, \tag{2.3}$$

where $u_{i,K} \equiv l^{-2}\int u_i(f(y))\,K(y)\,\mathrm{d}y$, Q_i is the available kinetic energy of component i within the eddy, ρ_0 the constant density, and $\alpha \in [0, 1]$ the parameter setting the fraction of extractable energy transferred towards isotropy. Solving the per-component energy balance for the amplitudes yields

$$c_i = \frac{27}{4l}\left[-u_{i,K} + \mathrm{sgn}(u_{i,K})\sqrt{u_{i,K}^2 + \alpha\sum_j \mathsf{T}_{ij}\,u_{j,K}^2}\,\right], \tag{2.4}$$

real for $0 \le \alpha \le 1$. The transfer matrix T equalises the component energies: the construction is a model of the return-to-isotropy (*slow*) part of the pressure–strain interaction, a correspondence made precise in § 2.2. The standard model contains no counterpart of the rapid, mean-strain-driven part.

The event sequence is a Poisson process whose rate density $\lambda(y_0, l; t)$ is set by the instantaneous flow state through the standard eddy time-scale construction, with a rate constant, a viscous cutoff and a large-eddy suppression parameter as the only tunable constants (Kerstein 1999; Stephens & Lignell 2021); the inertial-range cascade and the $k^{-5/3}$ spectrum are *outcomes* of repeated sampling from this rate, not inputs (Kerstein 1999). Finally, one diagnostic caveat of Kerstein *et al.* (2001) is respected throughout: because the line velocity does not itself advect fluid, off-diagonal products of line velocity components are not momentum fluxes; diagonal component spectra and variances are read from the fields, whereas shear stresses are read from eddy-induced transport.

2.2. *Rapid and slow pressure–strain, and linear rapid distortion*

The Reynolds-stress transport equation for a fluctuating field about a mean with constant gradient A_{ij} reads, in homogeneous turbulence,

$$\frac{\mathrm{d}R_{ij}}{\mathrm{d}t} = \mathcal{P}_{ij} + \Pi_{ij} - \varepsilon_{ij}, \qquad \mathcal{P}_{ij} = -A_{ik}R_{kj} - A_{jk}R_{ki}, \qquad \Pi_{ij} = \left\langle \tfrac{p'}{\rho}\left(\partial_j u_i' + \partial_i u_j'\right)\right\rangle, \tag{2.5}$$

with ε_{ij} the dissipation tensor (Pope 2000). The fluctuating pressure satisfies a Poisson equation whose source splits into a part linear in the mean gradient and a part quadratic in the fluctuations (Pope 2000),

$$\frac{1}{\rho}\nabla^2 p' = \underbrace{-2\,A_{lm}\,\partial_l u_m'}_{\text{rapid}} \;-\; \underbrace{\partial_l\partial_m\left(u_l'u_m' - \langle u_l'u_m'\rangle\right)}_{\text{slow}}, \tag{2.6}$$

and the pressure–strain correlation inherits the decomposition $\Pi_{ij} = \Pi_{ij}^{(\mathrm{r})} + \Pi_{ij}^{(\mathrm{s})}$ into a *rapid* part driven by the mean strain and a *slow* part driven by the turbulence alone. The slow part relaxes the Reynolds stress towards isotropy—the mechanism that the kernel (2.3)–(2.4) reproduces in model form. The rapid part is linear in A_{lm} and is determined by the velocity spectrum tensor $\Phi_{ij}(\boldsymbol{\kappa})$; the standard model represents neither $\mathcal{P}_{ij}$ nor $\Pi_{ij}^{(\mathrm{r})}$.

When the strain outruns the eddy turnover—the rapid limit—the response is computed exactly by linear theory (Batchelor & Proudman 1954; Hunt 1973; Hunt & Carruthers 1990). Writing the fluctuation as Fourier modes that follow the mean flow, each mode of time-dependent wavevector $\boldsymbol{\kappa}(t)$ and amplitude $\hat{u}_i$ evolves, in the inviscid rapid limit, according to

$$\frac{\mathrm{d}\kappa_i}{\mathrm{d}t} = -A_{ji}\,\kappa_j, \qquad \frac{\mathrm{d}\hat{u}_i}{\mathrm{d}t} = \left(\frac{2\,\kappa_i\kappa_m}{\kappa^2} - \delta_{im}\right) A_{mn}\,\hat{u}_n. \tag{2.7}$$

The bracketed operator combines the production $-A_{im}\hat{u}_m$ with the rapid pressure response, the solenoidal projection $P_{ij} = \delta_{ij} - \kappa_i\kappa_j/\kappa^2$ applied to the strained field; the factor of two is fixed by continuity, since only that coefficient makes $\mathrm{d}(\kappa_i\hat{u}_i)/\mathrm{d}t$ vanish under (2.7). Two features are central to what follows. First, the pressure response is *non-local* and intrinsically *three-dimensional*: it requires all of $\kappa_1, \kappa_2, \kappa_3$, whereas a single line resolves only the wavenumber along its own coordinate, so (2.7) cannot be evaluated on the line as written. Second, the rapid/slow decomposition furnishes the organising principle of the formulation: the slow part is already carried by the kernel, and it remains to supply the production and a rapid redistribution consistent with (2.7), expressed in quantities available on the line.

2.3. *The strain-coupled line equation and its structural admissibility*

The production term of the fluctuation momentum equation, $-A_{ij}u_j$, is a pointwise linear combination of the local velocity components and is therefore exactly representable on the line. Between eddy events the molecular evolution is replaced by

$$\partial_t u_i = \nu\,\partial_y^2 u_i \;-\; A_{ij}\,u_j \;+\; B_{ij}\,u_j, \tag{2.8}$$

in which $-A_{ij}u_j$ is the production and $B_{ij}u_j$ is the rapid pressure–strain operator constructed in § 2.4; the eddy events (2.1)–(2.4) are retained without modification, and the advancement is split between the continuous phase (2.8) and the discrete event sequence.

The forcing must not violate the structural properties on which the model rests, and it does not; we record the argument as a consistency check. The measure preservation, scale locality and kernel identities (2.2) are geometric properties of the map f and kernel K on a fixed domain, and the operator splitting applies the discrete event exactly as in the standard model, so they hold verbatim. Continuity of the fields is preserved because, for a mean gradient uniform along the line, the operator $\mathcal{A}_{ij} \equiv -A_{ij} + B_{ij}$ acts pointwise and the solution of the non-diffusive part of (2.8) over a sub-step is

$$u_i(y,t) = \left[\exp\!\int_{t_0}^{t} \mathcal{A}\,\mathrm{d}t'\right]_{ij} u_j(y,t_0), \tag{2.9}$$

a y-independent matrix, which maps continuous profiles to continuous profiles while the molecular term is smoothing. The forcing displaces no fluid along y, so the Lebesgue measure of the domain is untouched; the deliberate straining of the domain is a separate operation (§ 2.5).

The energetic role of the forcing is exact. At fixed domain length the production term evolves the Reynolds stress as

$$\left.\frac{\mathrm{d}R_{ij}}{\mathrm{d}t}\right|_{\mathrm{prod}} = -A_{ik}R_{kj} - A_{jk}R_{ki} \;\equiv\; \mathcal{P}_{ij}, \tag{2.10}$$

exactly the Reynolds-stress production tensor of (2.5): the term carries no model assumption. The kinetic-energy budget of the formulation thus separates into an exact production by the mean strain, energy-conserving redistribution (the kernels), and dissipation by molecular diffusion—the structure of the exact transport equation. The forcing adds the one term the standard model lacks without disturbing the terms it already represents.

2.4. *The rapid redistribution operator*

Were the continuous evolution to consist of production and diffusion only ($B_{ij} = 0$), the line would reproduce $\mathrm{d}R_{ij}/\mathrm{d}t = \mathcal{P}_{ij}$, whereas the exact inviscid rapid-distortion balance is $\mathrm{d}R_{ij}/\mathrm{d}t = \mathcal{P}_{ij} + \Pi^{(\mathrm{r})}_{ij}$. The omission is not small: for turbulence isotropic at the onset of straining the classical result recovered below gives $\Pi^{(\mathrm{r})}_{ij} = -\frac{3}{5}\mathcal{P}_{ij}$, so production acting alone overstates the initial distortion rate by a factor of $\frac{5}{2}$; under sustained strain it also lets a component energy grow without the bound that incompressibility imposes. Representing $\Pi^{(\mathrm{r})}_{ij}$ is essential, not optional.

2.4.1. *The exact rapid pressure–strain and the closure problem*

The exact rapid term follows from the amplitude equation. Writing (2.7) as $\mathrm{d}\hat{u}_i/\mathrm{d}t = M_{in}\hat{u}_n$ with $M_{in} = -A_{in} + 2(\kappa_i\kappa_m/\kappa^2)A_{mn}$, and noting that the incompressible wavevector dynamics preserve the measure $\mathrm{d}\boldsymbol{\kappa}$ (their phase-space divergence is $-A_{ii} = 0$), the spectrum tensor $\Phi_{ij} = \langle \hat{u}_i^*\hat{u}_j\rangle$ yields

$$\frac{\mathrm{d}R_{ij}}{\mathrm{d}t} = \int \left(M_{in}\Phi_{nj} + M_{jn}\Phi_{in}\right)\mathrm{d}\boldsymbol{\kappa} = \mathcal{P}_{ij} + \Pi^{(\mathrm{r})}_{ij}, \tag{2.11}$$

in which the production separates from the pressure contribution,

$$\Pi^{(\mathrm{r})}_{ij} = 2A_{mn}\left(\mathcal{R}_{ijmn} + \mathcal{R}_{jimn}\right), \qquad \mathcal{R}_{ijmn} \equiv \int \frac{\kappa_i\kappa_m}{\kappa^2}\,\Phi_{nj}(\boldsymbol{\kappa})\,\mathrm{d}\boldsymbol{\kappa}. \tag{2.12}$$

The tensor is traceless—by solenoidality $\kappa_n\Phi_{nj} = 0$, the trace integrand contains $\kappa_i\Phi_{ni} = 0$—so the rapid term redistributes energy among components without changing the total, in keeping with the energy-conserving character of the kernel mechanism. Equation (2.12) also exposes the central difficulty: the rapid term is a functional of the spectrum tensor through the orientation-weighted integral $\mathcal{R}_{ijmn}$, not of the single-point Reynolds stress—the classical closure problem of the rapid pressure–strain, common to every single-point model (Crow 1968; Launder *et al.* 1975; Pope 2000) and intrinsic to the physics. A line carries R_{ij} and its own one-dimensional spectra, but not the distribution of energy over wavevector *orientation* that $\mathcal{R}_{ijmn}$ requires; a spectral-tensor closure is therefore unavoidable, and we make it explicit. (What a line's data *do* determine about this integral, and how sharply, is the analytical question taken up in § 5.)

2.4.2. *Baseline isotropic closure*

The minimal closure takes the spectrum tensor instantaneously isotropic, $\Phi_{nj} = \frac{E(\kappa)}{4\pi\kappa^2}(\delta_{nj} - \kappa_n\kappa_j/\kappa^2)$ with $\int E\,\mathrm{d}\kappa = k_t$. With the angular averages $\overline{e_i e_m} = \frac{1}{3}\delta_{im}$ and $\overline{e_i e_m e_n e_j} = \frac{1}{15}(\delta_{im}\delta_{nj} + \delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn})$ for $e_i = \kappa_i/\kappa$,

$$\mathcal{R}^{\mathrm{iso}}_{ijmn} = k_t\left[\tfrac{4}{15}\delta_{im}\delta_{nj} - \tfrac{1}{15}\big(\delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn}\big)\right], \tag{2.13}$$

and substitution into (2.12) gives, with $S_{ij} = \frac{1}{2}(A_{ij} + A_{ji})$ and $A_{nn} = 0$,

$$\Pi^{(\mathrm{r}),\mathrm{iso}}_{ij} = \tfrac{4}{5}\,k_t\,S_{ij} \;=\; -\tfrac{3}{5}\,\mathcal{P}_{ij} \quad (\text{isotropic } R_{ij}), \tag{2.14}$$

the exact response of initially isotropic turbulence to a weak rapid strain (Crow 1968) and the content of the isotropisation-of-production model with $C_2 = \frac{3}{5}$ (Pope 2000)—obtained here, with no adjustable constant, as the isotropic limit of (2.12). For anisotropic turbulence $\mathcal{R}_{ijmn}$ must be modelled as a tensor function of R_{ij}; the general representation linear in the anisotropy is that of Launder *et al.* (1975), to which the construction reduces in form. We adopt (2.13) as the baseline and assess the anisotropic correction in § 3.

2.4.3. *Realisation on the line*

Given the closed rapid term—symmetric and traceless—we require a pointwise operator B_{ij} whose action on the line reproduces it. Demanding that the operator evolve the Reynolds stress by exactly $\Pi^{(\mathrm{r})}_{ij}$,

$$\big(\mathsf{BR} + \mathsf{RB}\big)_{ij} = \Pi^{(\mathrm{r})}_{ij}, \tag{2.15}$$

a continuous Lyapunov equation with, for positive-definite R, a unique symmetric solution B. The solution conserves kinetic energy automatically—contracting (2.15) gives $2B_{ij}R_{ij} = \Pi^{(\mathrm{r})}_{ii} = 0$, so the rate of working of the operator on the line vanishes. In the isotropic case $\mathsf{R} = \frac{2}{3}k_t\mathsf{I}$ it reduces to the closed form

$$B_{ij} = \tfrac{3}{5}\,S_{ij}, \qquad \mathcal{A}_{ij} = -A_{ij} + \tfrac{3}{5}\,S_{ij}. \tag{2.16}$$

B_{ij} is evaluated from the running Reynolds stress and applied continuously, paced by the mean strain, exactly as the production is. This is deliberate: the rapid pressure–strain is by construction the part of the redistribution driven by the mean gradient and therefore acts at the strain rate, in contrast to the slow redistribution that the eddy kernel supplies at the eddy rate. Strain-driven effects (production and rapid pressure–strain) are continuous and enter through $\mathcal{A}_{ij}$; turbulence-driven effects (the inertial cascade and the slow return to isotropy) remain the discrete events.

The sense in which the formulation is rapid-distortion consistent can now be stated precisely, again as a consistency check rather than a theorem. By (2.10) the production term contributes $\mathcal{P}_{ij}$ to $\mathrm{d}R_{ij}/\mathrm{d}t$ and by construction (2.15) the operator contributes $\Pi^{(\mathrm{r})}_{ij}$; their sum is the exact balance (2.11) for the prescribed closure, with no adjustable constant, recovering (2.14) at the onset of distortion of isotropic turbulence. Two qualifications delimit this statement: the consistency holds at the level of the *component energies* and is *conditional on the spectral closure*; the isotropic closure is exact only at onset, and its accuracy for anisotropic states is that of the isotropisation-of-production approximation it reproduces. The *spectral* distribution of the distorted energy is not enforced by (2.8): the operator acts uniformly on every resolved wavenumber, and the distribution of energy across scales emerges from its interaction with diffusion, the cascade and the domain straining. It is therefore a genuine prediction, tested in §§ 3–4—where the uniform, wavenumber-blind action of B_{ij} is also identified as the model's measured spectral deficiency (§ 4.6.1).

2.5. *Wavevector kinematics by domain straining*

The second ingredient of (2.7), the wavevector evolution, carries the kinematic compression and stretching of scales and is independent of the amplitude evolution: in the method-of-characteristics reading of linear theory, the wavevectors are the characteristics along which the amplitudes evolve. On the line the relevant wavenumber is κ_2. We restrict attention to a line aligned with a principal axis of the mean strain, so that $A_{12} = A_{32} = 0$ and a material element of the line remains a coordinate line—as holds for the irrotational mean flow of a stagnation region. Then $\mathrm{d}\kappa_2/\mathrm{d}t = -A_{22}\,\kappa_2$, and since a material element of length ℓ stretches as $\dot{\ell}/\ell = A_{22}$ while wavenumbers scale as $1/\ell$, a uniform dilatation of the domain reproduces the wavevector kinematics exactly:

$$\frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t} = A_{22}, \qquad \Longrightarrow \qquad \kappa_2(t) = \kappa_2(t_0)\,\exp\Bigl(-\int_{t_0}^{t} A_{22}\,\mathrm{d}t'\Bigr), \tag{2.17}$$

with L the domain length, implemented within the Lagrangian adaptive-mesh advancement of the model (Lignell *et al.* 2013) by imposing the mean-flow stretching on the mesh.

This step is not redundant with the amplitude operator. The Reynolds stress is an intensive second moment and is invariant under the uniform dilatation: the dilatation relabels the scales that carry the energy without changing the component energies, while $\mathcal{A}_{ij}$ changes the component energies without relabelling scales. The two operations are the physical-space counterparts of the characteristic motion and the amplitude evolution in (2.7), and together they account for the mean straining without double counting. Two limitations are inherent to representing a three-dimensional kinematics on one axis. First, the transverse wavenumbers also strain but are not represented; their effect is subsumed in the spectral closure, which already supplies the orientation information the line lacks. Second, a rotational mean flow, or a line not aligned with a principal axis, tilts the material line out of the coordinate direction and lies outside the formulation.

2.6. *Summary of the formulation*

The strain-coupled model advances the line field over a sub-step by composing (a) a *continuous strain–diffusion phase on a dilating domain*—integrate $\partial_t u_i = \nu\partial_y^2 u_i + \mathcal{A}_{ij}u_j$ with $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$, the production exact and B_{ij} given by (2.15) from the running Reynolds stress and the spectral closure (baseline (2.16)), while dilating the domain by (2.17)—and (b) *discrete eddy events* sampled from the standard rate distribution: the triplet map and the slow kernel, unchanged. Step (a) carries the mean-strain physics; step (b) carries the turbulence-driven physics; standard ODT is recovered exactly when $A_{ij} = 0$.

The relative importance of the two steps is governed by the ratio of the mean strain rate $S \equiv (2S_{ij}S_{ij})^{1/2}$ to the turbulence relaxation rate,

$$\chi \equiv \frac{S\,k_t}{\varepsilon}. \tag{2.18}$$

For $\chi \gg 1$ the distortion accumulates faster than the turbulence can respond, the eddy events are negligible over the distortion time, and the formulation reduces to the rapid-distortion limit of step (a), exact for the component energies under the closure. For $\chi \ll 1$ the turbulence relaxes between increments of strain and the slow kernel dominates. The stagnation region of a leading edge is the intermediate regime $\chi = O(1)$, in which neither limit is uniformly valid and both mechanisms contribute; the fidelity of the formulation across this range, and of the emergent distorted spectrum, are assessed against rapid-distortion solutions and scale-resolving data in §§ 3–4.

3. Verification, and moment-level validation

We separate two questions. *Verification* asks whether the formulation and its implementation reproduce a known exact solution in the regime where one exists—the rapid-distortion limit, tested analytically and in the solver below. *Validation* asks whether the model reproduces the behaviour of real strained turbulence outside that limit; the spectral validation, against the strained-box DNS with its exact-RDT companion, is the subject of § 4, and this section closes with the moment-level validation against the strained-turbulence DNS of Lee & Reynolds (1985). Every test in this section operates at the Reynolds-stress level, by design; no spectral claim rests on it, and none is made before the spectral benchmarks of § 4.

One point of identification, to be kept in view throughout this section and the next. The model under test is the strain-coupled formulation (SC-ODT); *standard* ODT is its $A_{ij} = 0$ limit (§ 2.6) and, having no mean-strain coupling—neither the production forcing, the rapid pressure–strain operator, nor the domain dilatation—produces none of the strained curves shown here. Every departure from isotropy in what follows is generated by the added strain coupling, not by the eddy dynamics of the standard model, which supplies only the cascade and the slow return to isotropy. Where the figures write "SC-ODT" they mean this formulation; "standard ODT" is reserved for the baseline, and the relaxation mechanism of § 6 ("clock-relaxed ODT") does not enter until Part II.

3.1. *Canonical strains and the exact onset rate*

Two irrotational homogeneous strains admit exact rapid-distortion solutions and isolate the physics of interest. *Plane strain*,

$$\mathsf{A} = \mathrm{diag}(a, -a, 0), \qquad a > 0, \tag{3.1}$$

is the strain of a two-dimensional stagnation region: with x_1 tangential to the surface, x_2 normal to it and x_3 spanwise, the component compressed by the strain is the wall-normal (upwash) component u_2, whose amplification is the pre-impact distortion relevant to leading-edge noise. *Axisymmetric strain*,

$$\mathsf{A} = \mathrm{diag}(-\tfrac{1}{2}\gamma, -\tfrac{1}{2}\gamma, \gamma), \qquad \gamma > 0, \tag{3.2}$$

is the strain of a contraction or a three-dimensional stagnation flow and provides an independent test in which two components are amplified and one suppressed. For both, A is symmetric, so the principal-axis alignment assumed in § 2.5 holds; the line is taken along x_2.

For turbulence isotropic at the onset of straining the continuous evolution (2.8) gives, by (2.10) and (2.14),

$$\left.\frac{\mathrm{d}R_{ij}}{\mathrm{d}t}\right|_0 = \mathcal{P}_{ij} + \Pi^{(\mathrm{r}),\mathrm{iso}}_{ij} = \tfrac{2}{5}\,\mathcal{P}_{ij} = -\tfrac{8}{15}\,k_t\,S_{ij}, \tag{3.3}$$

the exact rapid-distortion rate for initially isotropic turbulence, with no adjustable constant. This is the sharpest verification available: a closed-form prediction the model must satisfy identically, not approximately. Evaluated for the two canonical strains it gives, under plane strain, onset rates $\mp\frac{8}{15}k_t a$ for the stretched and compressed components with the neutral component unchanged, and, under axisymmetric strain, $+\frac{4}{15}k_t\gamma$ for each lateral component against $-\frac{8}{15}k_t\gamma$ for the axis. For plane strain the predicted upwash growth rate $+\frac{8}{15}k_t a$ is to be compared with the $\frac{4}{3}k_t a$ that production alone would give—the factor-$\frac{5}{2}$ overstatement anticipated in § 2.4. A model that passes (3.3) for both strains has its production and rapid terms verified jointly; failure would indicate an error in the operator $\mathcal{A}_{ij}$ or its implementation, isolated from any closure or cascade effect.

3.2. *Finite total strain, and the necessity of the rapid term*

Beyond onset the turbulence becomes anisotropic and the rapid term (2.12) ceases to be a function of the Reynolds stress alone, so the verification becomes a test of the spectral closure rather than of the construction. We integrate the model moment equation along each canonical strain with two closures: the baseline isotropic closure (2.13) (equivalently, isotropisation of production with $C_2 = 3/5$; IP) and the quasi-isotropic model of Launder *et al.* (1975) (LRR),

$$\Pi^{(\mathrm{r})}_{ij} = C_2\,k_t S_{ij} + C_3\,k_t\left(b_{ik}S_{jk} + b_{jk}S_{ik} - \tfrac{2}{3}b_{mn}S_{mn}\delta_{ij}\right) + C_4\,k_t\left(b_{ik}W_{jk} + b_{jk}W_{ik}\right), \tag{3.4}$$

with anisotropy $b_{ij} = R_{ij}/2k_t - \frac{1}{3}\delta_{ij}$, mean rotation $W_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$ (zero for the irrotational strains considered), and $C_2 = 4/5$, $C_3 = 7/4$, $C_4 = 131/100$; the value $C_2 = 4/5$ is fixed by requiring that (3.4) reduce at isotropy to the exact $\frac{4}{5}k_t S_{ij}$ of (2.14), so LRR shares the constant-free onset slope. The benchmark is the exact rapid-distortion solution, obtained by integrating the spectral equations (2.7) over a Gauss–Legendre ensemble of initial wavevector directions on the unit sphere: for initially isotropic turbulence the modal dynamics depend only on direction, so the Reynolds stress is the spherical average of the evolved modal covariance. This integration is independent of the moment models and reproduces the onset slope (3.3) to within 10^{-16}, verifying the rapid-distortion machinery itself.

Figure 1 reports the component-energy fractions against total strain. Three observations follow. First, at onset all curves for each component are tangent—the constant-free identity (3.3) is satisfied—and through $e \lesssim 1$ the closures are nearly indistinguishable from the exact solution. Second, as anisotropy accumulates the closures depart, LRR uniformly less than IP: for axisymmetric strain at $e = 4$ the lateral fraction is 0.498 (exact), 0.468 (LRR), 0.450 (IP), and the axial fraction 0.003, 0.063, 0.100. This gap is the spectral-closure error in isolation—the eddy events are inactive in the rapid limit—and it bounds the accuracy attainable from any single rapid operator. Third, and most important to state plainly, for plane strain both closures miss a *qualitative* feature: in the exact solution the neutral spanwise component grows monotonically, overtakes the upwash near $e \approx 3.3$, and is the largest component by $e = 4$ ($\overline{u_3^2}/2k_t = 0.51$ against 0.47), whereas both closures predict it to decay. This redistribution into the neutral direction is governed by the orientation distribution of the spectral tensor and cannot be recovered from single-point quantities; it

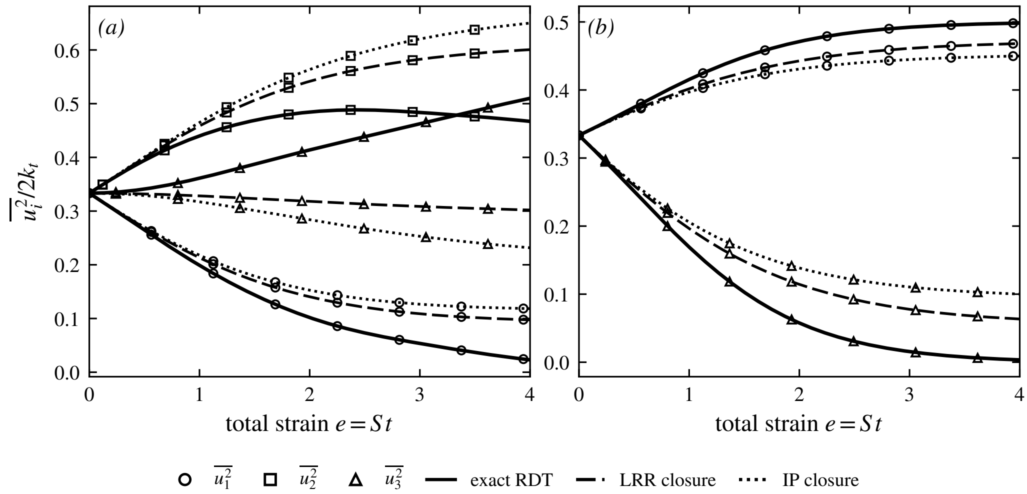

Figure 1. Component energy fractions against total strain for initially isotropic turbulence under (*a*) plane strain and (*b*) axisymmetric strain, normalised to $S = 1$: exact rapid-distortion theory (wavevector-ensemble integration of (2.7)) against the SC-ODT moment equation with the LRR closure (3.4) and the IP closure (2.13). Line style denotes the source (solid exact RDT, dashed LRR, dotted IP) and marker the component ($\circ\, \overline{u_1^2}$, $\square\, \overline{u_2^2}$, $\triangle\, \overline{u_3^2}$). For axisymmetric strain the two lateral components coincide and only $\overline{u_1^2} = \overline{u_2^2}$ (lateral) and $\overline{u_3^2}$ (axial) are shown. All closures share the exact onset slope; LRR tracks the exact solution more closely at finite strain.

is the moment-level signature of precisely the transverse-wavenumber information a single line does not carry. Section 5.6 returns to it with the strained-box DNS in hand and localises the deficiency in the closure's b-dependent term.

The upwash component, whose spectrum is the input to the leading-edge response, makes the case for the rapid term quantitative (figure 2). Production acting alone drives the upwash fraction from $1/3$ to 0.98 by $e = 4$—almost the entire kinetic energy forced into one component, an essentially non-realisable state—and overstates the initial amplification rate by the factor $\frac{5}{2}$. The rapid term removes three fifths of this at onset and keeps the fraction physical: the exact solution peaks at 0.49 near $e \approx 2.4$ and then relaxes as energy passes to the spanwise component. Both closures reproduce the amplification itself—the effect that the frozen, isotropic input of classical leading-edge-noise prediction omits—while overpredicting it at large strain (LRR 0.60, IP 0.65 at $e = 4$, against the exact 0.47). The rapid pressure–strain is therefore not a refinement but a necessity, and the accuracy of the method for a given configuration is controlled by the total strain accumulated along the stagnation streamline.

3.3. *Verification of the implementation*

The preceding checks verify the *formulation*. They do not test the *implementation*, in which the operator is not prescribed but reconstructed at every substep: the line Reynolds stress is formed from the resolved field, the closure supplies $\Pi_{ij}^{(\mathrm{r})}$, the Lyapunov equation (2.15) is solved for B_{ij}, and $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$ is applied pointwise. To isolate this chain the solver is run in the rapid-distortion limit—plane strain, eddy events suppressed, zero molecular diffusion—from a statistically isotropic resolved initial field, and the component energies are evaluated as length-weighted line averages. With the events inactive the line's component-energy evolution must coincide with the closed moment equation across the whole strain range, not merely at onset.

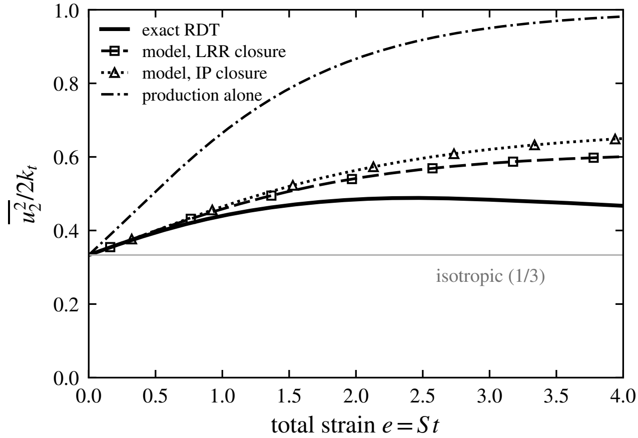

Figure 2. Upwash energy fraction under plane strain for initially isotropic turbulence: exact rapid-distortion theory, SC-ODT with the LRR and IP closures, and production alone ($B_{ij} = 0$). Production without the rapid term overstates the onset amplification rate by $\frac{5}{2}$ and forces almost all the energy into the upwash component; the rapid term restores the physical, bounded amplification.

Figure 3 shows that it does: the solver output lies on the LRR trajectory over $0 \leq e \leq 4$ to within 10^{-4} in each component fraction, so the per-substep stress reconstruction, Lyapunov solve and operator assembly are correct at every state of anisotropy. The comparison must be read correctly: the solver reproduces the *closure*, and it should—with the events suppressed the line carries no information beyond the single-point statistics the closure itself uses. The offset from the exact solution is the single-point closure error of § 3.2, not an implementation error; agreement with the exact solution here would indicate a fault. The fractions test only the anisotropy; the kinetic energy, which the traceless rapid term cannot change directly and which grows through the production integral $\mathrm{d}\ln k_t/\mathrm{d}e = f_{22} - f_{11}$, provides the independent magnitude check, and the solver reproduces the LRR amplification to within 0.3%. For definiteness: every strained computation in the remainder of the paper uses the LRR closure, whose error against the exact solution is quantified above at every strain reached; the sensitivity of the spectral conclusions to this closure choice is not estimated but measured, through the exact-RDT companions of § 4 and the axisymmetry measurement of § 5, which carry the closure deficiency into the quoted uncertainty rather than leaving it implicit. Reaching that figure required a resolved spectral initialisation (a white-noise start places energy at the grid scale, where mesh-adaption interpolation damps it) and a strain-resolved timestep bound in the inviscid limit; both effects act nearly isotropically, so the energy panel is what exposes them.

3.4. *Moment-level validation against the strained-turbulence DNS of Lee & Reynolds*

The DNS of Lee & Reynolds (1985) is retained as a benchmark for one specific reason: it was deliberately run at a turbulence Reynolds number low enough ($q^4/\nu\varepsilon \lesssim 100$) to be fully resolvable on a single line, and it reports the Reynolds-stress anisotropy against total strain for several irrotational strain modes alongside the analytic rapid-distortion prediction—a graded, moment-level reference that the spectral benchmarks of § 4 do not provide. The same programme's later rapid-limit analysis (Lee *et al.* 1986) supplies the analytic reference used in the supplementary CMK comparison; higher-Reynolds-number strained computations exist, but the spectral role they would serve is filled here by the dedicated strained-box DNS of § 4, leaving Lee & Reynolds's low-Reynolds-number study

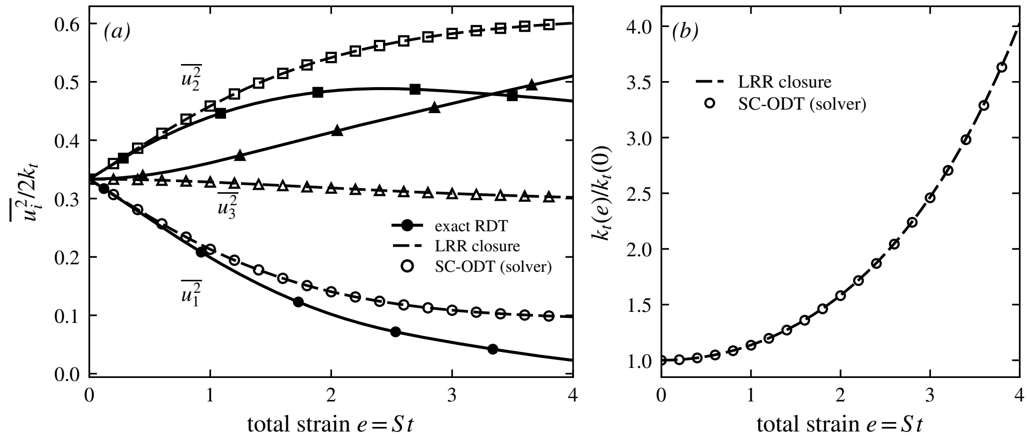

Figure 3. Verification of the SC-ODT implementation under plane strain, eddy events suppressed and zero molecular diffusion. (*a*) Component energy fractions; (*b*) kinetic-energy amplification $k_t(e)/k_t(0)$. Open symbols: the SC-ODT line solver (length-weighted line averages), by component ($\circ\,\overline{u_1^2}$, $\square\,\overline{u_2^2}$, $\triangle\,\overline{u_3^2}$); dashed: the LRR closure integrated as a moment equation; solid with filled symbols: exact rapid-distortion theory. The solver reproduces the closure to 10^{-4} in the fractions and 0.3% in the energy; the offset from the exact solution is the closure error of § 3.2, as it must be.

as the appropriate moment-level benchmark for a fully resolved line. The comparison quantity is the anisotropy b_{ij} against the deformation ratio $c = \exp\int S\,\mathrm{d}t$ (so $\ln c$ is the total strain e of the present notation), computed from the single-point component variances along the line, ensemble-averaged over 1000 realisations; no spectral transform is involved. Two of their strain modes are run: plane strain and axisymmetric contraction, the line along a compressed direction in each case, to the total strains of the DNS ($c = 4$ and 3.32); the measured line compression agrees with $\exp(A_{22}t_f)$ to better than 0.3%, confirming the kinematic fidelity of the dilatation. One parameter distinction matters for interpretation: Lee & Reynolds characterise each run by $S^* = Sq^2/\varepsilon$, which equals 2χ identically since $q^2 = 2k_t$—a correspondence of definitions, not a calibration—and find the axisymmetric anisotropy essentially independent of S^*—a clean, parameter-free target—whereas in plane strain the unstrained component b_{11} is strongly S^*-dependent, spanning $b_{11} \approx 0$ (slow) to $+0.12$ (rapid) in the DNS.

Under plane strain (figure 4) the model reproduces the two strained components quantitatively across the whole range: $b_{22} = +0.21$ at $c = 3.86$ against $\approx +0.20$ in the DNS, and $b_{33} = -0.20$ against ≈ -0.26, capturing the trend and the bulk of the magnitude. These are exactly the components the DNS identifies as strain-rate-independent, so they constitute the robust part of the comparison. Under axisymmetric contraction the model reproduces the sign structure and the monotonic approach to a strained state while under-predicting the anisotropy magnitude by roughly 25% ($b_{11} = -0.21$ against ≈ -0.29 at $c = 3.16$), dominated by the spectral-closure gap isolated in § 3.2 and common to the whole second-moment family; the line adds only a small artefact, breaking the transverse symmetry with a residual split $b_{22} = +0.102$ against $b_{33} = +0.113$ that the symmetry-respecting mean $\frac{1}{2}(b_{22} + b_{33})$ removes.

The one component the model does not reproduce, the plane-strain unstrained anisotropy b_{11}, is not a deficiency specific to the model, and figure 5 locates it exactly. Exact linear rapid distortion, evaluated by the wavevector-ensemble integration of § 3.2, gives b_{11} *rising* to $+0.114$ at $c = 4$: although $A_{1k} = 0$, the rapid pressure–strain feeds the neutral direction—the same orientation-carried redistribution as the spanwise growth in § 3.2. (A tempting

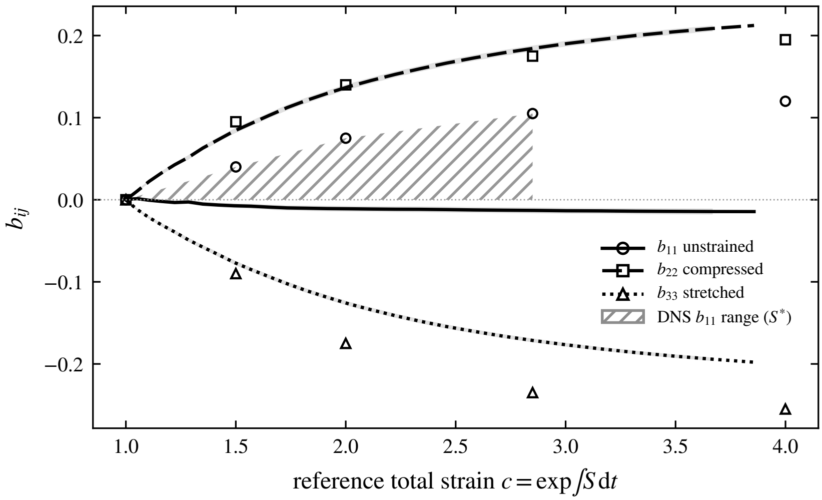

Figure 4. Reynolds-stress anisotropy under plane strain: SC-ODT (lines; ensemble mean over 1000 realisations, standard-error band) against the DNS of Lee & Reynolds (1985) (open symbols, digitised from their figure 5.4). Line style and marker denote the component (b_{11} solid/○, b_{22} dashed/□, b_{33} dotted/△); the hatched band is the DNS b_{11} range over slow-to-fast S^*. The strained components b_{22} and b_{33} are reproduced quantitatively; the unstrained b_{11} is strain-rate-dependent in the DNS, with SC-ODT at its slow-strain edge.

kinematic argument, that $\langle u_1^2 \rangle$ is conserved because the component is unstrained, holds for production but not for the rapid term, and gives the wrong sign; an earlier version of this comparison used it.) The fast-strain DNS therefore *tracks exact RDT*, as Lee & Reynolds themselves observed, while both single-point closures give $b_{11} \approx 0$, slightly negative: the closure family misses the neutral-direction feed, which is carried by the orientation distribution that single-point models discard—the same deficiency measured spectrally, and localised in the closure's b-dependent term, in § 5.6. SC-ODT coincides with the closure on which its operator is built, so its near-zero b_{11} is correct for its formulation and matches the DNS at slow strain rate, where the nonlinear scrambling erases the neutral-direction feed; at fast strain it inherits the closure family's known limitation rather than any line-specific artefact. The components that govern the upwash spectrum fed to the leading-edge response are precisely those the model reproduces quantitatively. Finally, Lee & Reynolds's own transport-budget analysis finds that in plane strain the intercomponent transfer is carried almost entirely by the rapid term—independent support for the rapid/slow decomposition on which § 5 rests.

4. The linear reference and the distorted spectrum

This section establishes what SC-ODT predicts for the strained spectrum and measures it against the correct linear reference and against DNS. It first separates the model's own kinematics—a rigid spectral translation—from the prediction of linear theory for the line-projected spectrum, which is *not* rigid (§§ 4.1–4.2); it then examines the emergent distorted spectrum and fixes the physical operating point (§§ 4.3–4.4); and it closes with the benchmark that meets the model-versus-itself objection head on—a strained-box DNS with an exact rapid-distortion companion, reduced to the same one-dimensional observables (§ 4.6).

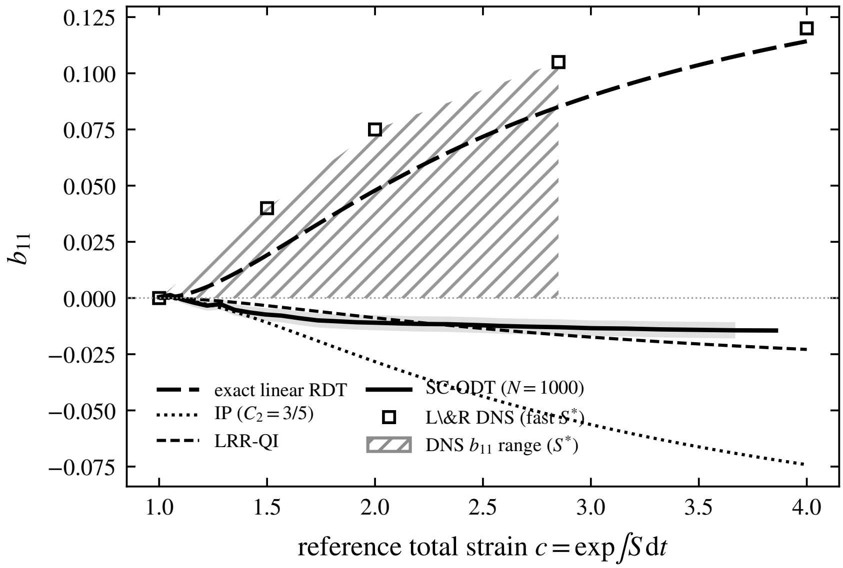

Figure 5. The unstrained-direction anisotropy b_{11} under plane strain: SC-ODT (solid, 1000 realisations, standard-error band) against exact linear rapid distortion (long-dashed, wavevector-ensemble integration), the IP (dotted) and LRR-QI (short-dashed) closures, and the DNS of Lee & Reynolds (1985) (open squares; hatched band = DNS range over slow-to-fast S^*). The fast-strain DNS tracks exact RDT, whose rapid pressure–strain feeds the neutral direction; the single-point closures—and SC-ODT, built on them—miss this orientation-carried redistribution, the deficiency measured and localised in § 5.6.

4.1. *Compression without cascade: the model's kinematic reference*

With the eddy events suppressed and zero molecular viscosity, the mean strain is the only process acting on the resolved field, and its effect on the line is purely kinematic. The line length contracts as $\mathcal{L}(e) = \mathcal{L}_0\, e^{A_{22}e}$, so every resolved wavenumber is rescaled as $k(e) = k(0)\, e^{-A_{22}e}$—the wavevector kinematics it shares with rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990)—while the continuous redistribution operator of § 2.4 acts uniformly on every resolved wavenumber. The combination transfers no energy between scales: a rigid translation of the line spectrum towards higher wavenumber. We emphasise at the outset that this rigid translation is a property of the SC-ODT *line*—uniform dilatation composed with a wavenumber-uniform operator—and not of linear distortion theory itself; the correct linear reference for the projected spectrum is constructed in § 4.2 below. As a check on the implementation, however, the rigid translation is an exact, parameter-free target.

Figure 7 confirms it. The component spectra translate toward higher k without change of shape—the peak neither broadens nor develops an inertial range—and the spectral centroid follows $\mathrm{e}^{-A_{22}e}$, reaching 7.03 at $e = 3.9$ against the exact 7.03, with the three components migrating identically; the domain length follows $\mathrm{e}^{A_{22}e}$ to four digits. The single-point moments are unchanged from the no-dilatation case, since a uniform rescaling of position leaves the length-weighted velocity statistics invariant. Compression therefore acts entirely on the spectrum and reproduces the model's kinematic prediction exactly. It is the kinematic reference (the dashed line in all centroid figures below) against which the full model is measured.

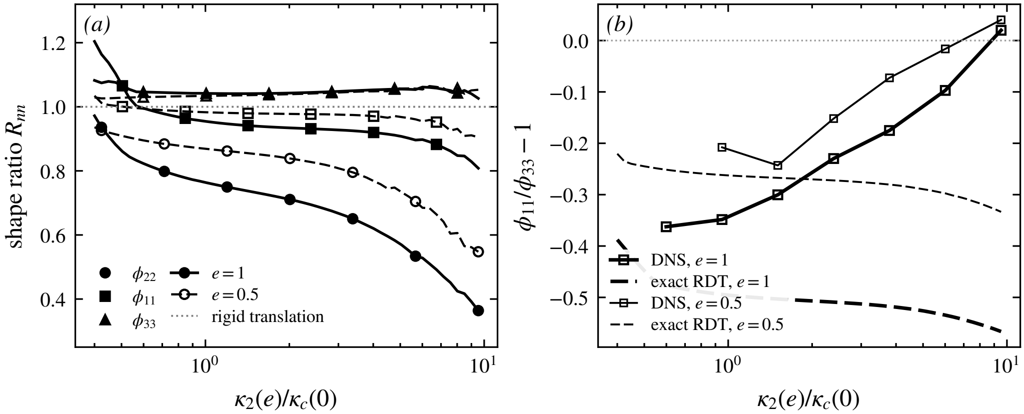

Figure 6. Exact linear theory does not translate the projected spectra rigidly. (*a*) Shape ratio $R_{nn}(\kappa_2)$ of the exactly distorted, line-projected component spectra to a rigid translation under plane strain, at $e = 1$ (solid, filled markers) and $e = 0.5$ (dashed, open markers); markers distinguish the components, and the dotted line is the rigid translation $R_{nn} = 1$. Closed-form Cauchy solution for an isotropic initial spectrum, integrated over the transverse wavenumber plane. (*b*) Transverse splitting $\phi_{11}/\phi_{33} - 1$ versus wavenumber in exact linear theory (dashed) and in the strained-box DNS of § 4.6 (solid, squares), at $e = 1$ (heavy) and $e = 0.5$ (light).

4.2. *The linear reference for the projected spectrum: exact RDT*

The rigid translation above must not be mistaken for the prediction of linear rapid-distortion theory for the quantity the line actually carries. In the full linear theory the amplitude of each Fourier mode evolves at a rate that depends on the orientation of its three-dimensional wavevector (equation 2.7); the one-dimensional component spectra are projections—integrals over the transverse wavenumber plane—and there is no reason for a projection of orientation-dependent amplification to translate rigidly. Whether it does is a question of calculation, and the answer is that it does not. Using the closed-form Cauchy solution of the linear problem for an isotropic initial spectrum, integrated analytically over the transverse wavenumber plane and band-limited to the wavenumbers resolved by the DNS box of § 4.6, we define the shape ratio $R_{nn}(\kappa_2)$ as the projected component spectrum at strain e, normalised by its integral, divided by the rigidly translated initial spectrum normalised likewise; $R_{nn} \equiv 1$ at every κ_2 if and only if the translation is rigid. At $e = 1$ under plane strain the upwash ratio R_{22} falls monotonically from 0.86 at half the initial spectral centroid to 0.36 at ten times it: exact linear theory redistributes the projected upwash spectrum towards low wavenumber by a factor of 2.4 across the resolved range, because modes aligned with the line ($\kappa \to \kappa_2$) are amplified least. The transverse components are affected more weakly (R_{11} from 1.07 to 0.81; R_{33} within seven per cent of unity), so the projected anisotropy of linear theory is itself scale-dependent. Figure 6 shows the ratios. The exact-RDT companions of the DNS (§ 4.6), which share its initial fields, reproduce these analytic curves to within 0.05 in the transverse splitting over the resolved band; the analytic integral is used because the companions sample the largest scales with one Fourier line per band.

Two consequences run through the remainder of the paper. First, the comparison of the full model against the rigid translation (§ 4.3) measures the departure of the model from its own kinematics; the departure of the *flow* from linear theory must be measured against the exact projected reference, and the two differ already at the linear level. Second, the model's rigid spectral kinematics is itself a structural approximation—linear theory deepens the projected anisotropy with wavenumber, which a wavenumber-uniform operator cannot

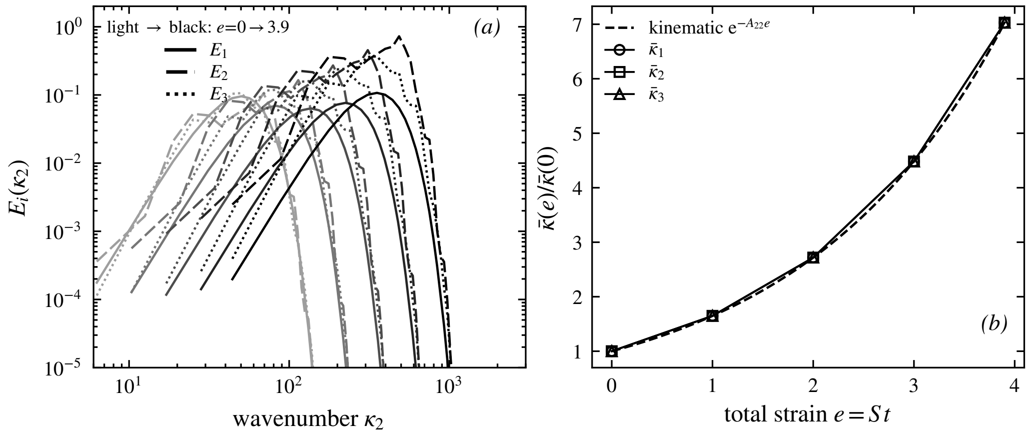

Figure 7. Compression without cascade (eddy events suppressed, inviscid). (*a*) Component line spectra $E_i(\kappa_2)$ at total strains $e = 0, 1, 2, 3, 3.9$ (lighter to darker); the spectra translate to higher wavenumber without changing shape. (*b*) Spectral-centroid migration $\bar{\kappa}(e)/\bar{\kappa}(0)$ for each component (symbols) against the kinematic prediction $e^{-A_{22}e}$ (dashed). The centroid follows the prediction and the three components coincide, confirming purely geometric compression of the line.

represent—and § 4.6.1 quantifies the resulting deficiency against both the exact reference and DNS.

4.3. *The distorted spectrum: strain on against strain off*

The central model-internal comparison isolates what the strain does to the spectrum at fixed Reynolds number, holding the turbulence otherwise identical. Two ensembles of 128 realisations are run at $\nu = 10^{-5}$: SC-ODT (eddies and dilatation active) and a baseline identical in every respect except that the strain machinery is switched off, so the line does not compress—which is standard ODT, the $A_{ij} = 0$ limit of § 2.6. Because the two differ only in the presence of the mean strain, the difference between them is the distortion, cleanly separated from the cascade and dissipation the eddies produce on their own.

Figure 8 shows the result, and it departs from the kinematic reference in a specific and physically meaningful way. Consider the centroid first. The no-strain baseline, after the brief initialisation transient, relaxes monotonically to $\bar{\kappa}/\bar{\kappa}_0 < 1$: with no compression, the eddies merely redistribute energy and the centroid drifts below its initial value. SC-ODT is lifted clearly above this baseline—compression does push the spectrum towards higher wavenumber—yet it settles at $\bar{\kappa}/\bar{\kappa}_0 \approx 1.6$–$2.1$, far below the rigid-translation line, which rises to 7.4 at $e = 4$. The reason is visible in the spectra: the SC-ODT spectrum is not a rigid translation of the baseline but a *broadband* distortion. It extends to substantially higher wavenumber than the baseline—energy is genuinely carried to small scales by compression and cascade—but the bulk of the energy remains at the energetic scales, and what reaches high wavenumber is dissipated there rather than accumulated.

Two distinct effects are mixed in this departure, and § 4.2 requires that they be separated. The first is the eddy cascade and dissipation, which the strain-on/strain-off construction isolates cleanly at the level of the model. The second is that exact linear theory itself, once projected onto the line, is already broadband (figure 6): part of what a rigid-translation reference attributes to nonlinearity is present in the linear physics. The strain-on/strain-off comparison therefore measures what the mean strain does *within the model*; how the model's distorted spectrum stands against exact linear theory and against DNS on the same footing is answered by the three-way comparison of § 4.6.1. What survives either

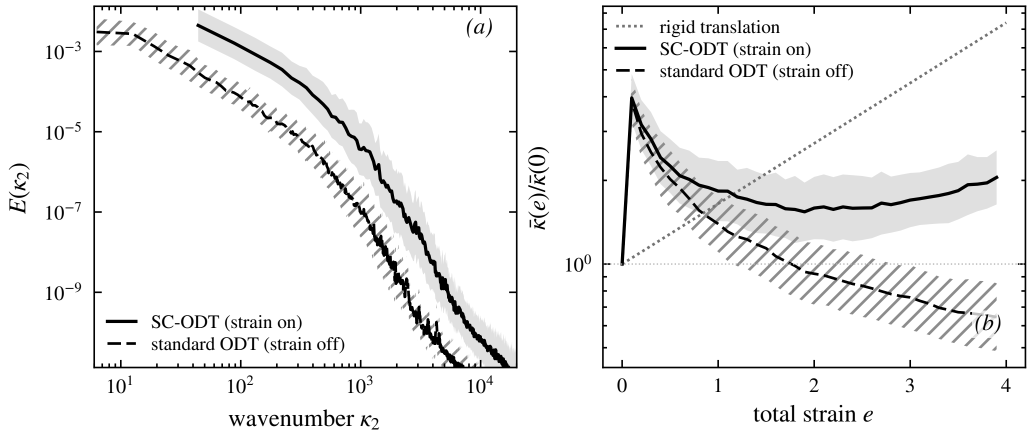

Figure 8. SC-ODT (strain on) against the matched strain-off baseline, standard ODT, at $\nu = 10^{-5}$ (ensemble means over 128 realisations, geometric $\pm\sigma$ bands). (*a*) Final-strain total spectrum $E(\kappa_2)$: SC-ODT (strain on) extends to higher wavenumber than the baseline (strain off) but remains broadband, not a rigid translation. (*b*) Centroid migration: the baseline relaxes below unity, SC-ODT is lifted above it by the compression, yet stays far below the rigid-translation line (dashed). The gap between the two solid curves is the distortion attributable to the mean strain at this Reynolds number.

way is the qualitative point that a frozen, rigidly strained upwash spectrum misrepresents the distorted inflow, which motivates supplying a distorted spectrum to Amiet's theory in place of it, in the spirit of the rapid-distortion-modified inflow spectra of de Santana *et al.* (2016) but without the frozen-shape assumption.

4.4. *The physical operating point*

The molecular viscosity is, in these units, the inverse integral-scale Reynolds number, and it is a physical property of the represented inflow, not a tunable parameter: a sweep over $\nu = 10^{-4}$–10^{-7} shows the cascade deepening monotonically, with the resolved cases settling below the linear-RDT centroid line while the lowest viscosities climb towards it only by piling energy into an under-resolved grid-scale shelf. The operating point is therefore the viscosity matching the target inflow, resolution-checked. For the reference configuration—a NACA0012 at $U_\infty = 20\,\mathrm{m\,s^{-1}}$ in grid turbulence with intensity $u'/U_\infty \approx 6\%$ and integral scale $\mathcal{L} \approx 55$ mm at the leading-edge plane—the integral-scale Reynolds number is $Re_{\mathcal{L}} \approx 4.4\times10^3$, which maps through the initial-condition normalisation to $\nu \approx 3\times10^{-6}$, run on a refined grid (16000 cells) on which the dissipation rolloff is resolved with margin; the ensemble comprises 200 realisations.

Figure 9 shows the result. The component spectra fall smoothly across four decades with no grid-scale shelf, the upwash component E_2 amplified above the streamwise E_1—the spectral signature of the plane strain—and no extended $k^{-5/3}$ range, as expected physically at this Reynolds number (Kolmogorov 1991; Pope 2000). The centroid settles after the initialisation transient and migrates to ≈ 2.4 by $e = 4$, far below the rigid-translation value of 7.4: the broadband, dissipative distortion of § 4.3 at the physical operating point. The regime is quantified with an estimator-free dissipation: because the eddy events conserve energy and the production is exact, molecular dissipation is the only sink, so it is fixed by the resolved energy budget

$$\varepsilon = \mathcal{P} - \frac{\mathrm{d}k_t}{\mathrm{d}t} = -A_{ij}R_{ij} - \frac{\mathrm{d}k_t}{\mathrm{d}t}, \tag{4.1}$$

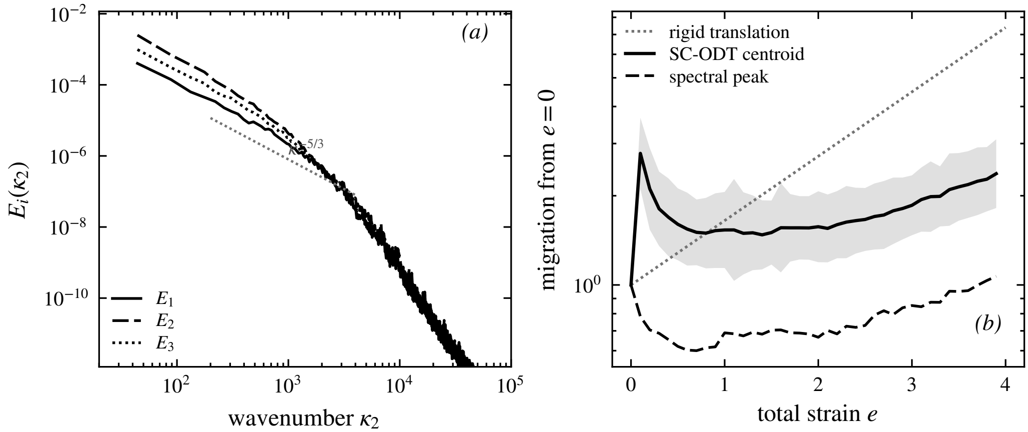

Figure 9. The physical operating point ($\nu \approx 3 \times 10^{-6}$, $Re_{\mathcal{L}} \approx 4.4 \times 10^3$, 16000 cells; ensemble over 200 realisations, geometric $\pm\sigma$). (*a*) Final-strain component spectra (solid E_1, dashed E_2, dotted E_3): smooth rolloff with no grid-scale shelf; the upwash E_2 amplified above the streamwise E_1; grey dotted reference slope $\kappa^{-5/3}$. (*b*) Migration of the total-spectrum centroid (solid) and the spectral peak (dashed) against the rigid-translation law (grey dotted): the broadband distortion that linear theory cannot represent, at the operating point and with the noise averaged out.

every term available on the line. By this budget the strain-rate parameter settles at $\chi = Sk_t/\varepsilon \approx 1.2$ over the strained trajectory (ensemble of 200 realisations), corroborated within a few tens of per cent by the direct gradient estimate $\langle 2\nu(\partial u_i/\partial y)^2\rangle \approx 1.1$: the operating point sits at $\chi = O(1)$, bracketed by the two strained-box DNS rapidities $Sk_t/\varepsilon = 0.8$ and 16 used in § 4.6.1, the regime the leading-edge stagnation flow occupies, where neither the rapid-distortion nor the near-equilibrium limit applies and where beating linear theory is meaningful. The spectral *peak* corroborates the regime reading independently of the dissipation estimator: the energy-containing wavenumber itself migrates far less than the rigid law, so the eddy relaxation holds back the energy-containing range, not merely the dissipation tail—linear theory misplaces the energy-containing wavenumber, not only the spectral shape at high κ. The cost is the practical justification of the approach: the operating-point ensemble completes in a few hundred core-hours, roughly two orders of magnitude below a strained DNS ensemble at comparable Reynolds number, which is what makes parametric sweeps over inflow condition feasible.

4.5. *Validation against the strained-turbulence experiment of Chen, Meneveau & Katz*

The strain-on/strain-off comparison is model-internal; an external benchmark at laboratory Reynolds number is provided by the plane-strain experiment of Chen *et al.* (2006) (CMK), whose straining phase (total deformation $D \approx 3$, $e \approx 2.20$ in the present notation) we reproduce. In the rapid-distortion limit (eddy events suppressed, inviscid) the model follows the analytic reference of Lee *et al.* (1986) that CMK use—component centroids on the rigid-translation law with deformation ratio 3.00 at $e = 2.20$, kinetic-energy amplification 1.72 against the exact 1.69, the residual being the moment closure's, not the implementation's (§ 3.3). With eddies and viscosity active the component centroids diverge under strain—streamwise migration 1.9 against 1.5 for the upwash at the end of the straining phase, both far below the rigid value 3, the upwash held back most—which is the qualitative departure-from-RDT behaviour CMK report and the spectral signature of the broadband distortion of § 4.3. The comparison is qualitative by construction (initial spectrum, spectral projection and scale separation all differ from the experiment's); the full account, with both figures

$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c:	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	−0.020	−0.041	−0.050	−0.054	−0.056
DNS	+0.032	−0.033	−0.029	−0.032	−0.036	−0.040	−0.046
SC-ODT	+0.103	+0.103	+0.096	+0.073	+0.075	+0.079	+0.076

Table 1. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. Bands below $0.75\,\kappa_c$ rest on a few box modes and are indicative only.

and the stated limits, is given in the supplementary material (§ S1), and the quantitative spectral assessment is carried out against the strained-box DNS below, where projection and initial spectrum are matched by construction.

4.6. *Validation against a strained-box DNS with an exact-RDT companion*

The comparisons above are either model-internal (§ 4.3) or, where external, qualitative by construction (§ 4.5) or at the Reynolds-stress level (§ 3.4). The referent quantity for the leading-edge application is spectral, and the objection that the emergent-spectrum results compare the model only with itself must be met with an independent, projection-matched spectral benchmark under the same strain. To that end we performed direct numerical simulations of plane-strained homogeneous turbulence in a deforming-frame (Rogallo-type) pseudo-spectral formulation: a 128^3 box, decaying isotropic precursor, then plane strain $A = S\,\mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ applied at strain-rate parameters $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to total strain $e = 1$, four independent realisations ($\kappa_{\max}\eta \approx 0.76$ at $e = 1$: a pilot resolution, so small-scale statements below are qualitative). Alongside every DNS realisation runs an *exact-RDT companion*: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically to the DNS and to ODT. The companion is what § 4.2 used to construct the linear reference; here it completes a three-way comparison in which model, linear theory and DNS are reduced to the same one-dimensional observables. On the ODT side the ensembles use an unstrained precursor: the line is evolved as ordinary ODT until the initialisation transient has relaxed and the eddy population is statistically active, and the strain is switched on only then, so that the strained evolution starts from a physically conditioned state rather than from the synthetic initial condition. All ODT statistics in this subsection are medians over 1024 realisations per case with bootstrap confidence intervals; median statistics are used because the strained band energies are strongly intermittent across realisations, and ensemble means of spectral ratios are dominated by rare events at any practical sample size.

4.6.1. *Three-way spectral anisotropy: ODT, exact RDT, DNS*

The comparison quantity is the strain-induced change of the spectral component anisotropy, $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e, minus the same quantity evaluated in the system's own unstrained state with each mode mapped to its strained wavenumber. Referencing each system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the longitudinal–transverse structure of the isotropic line spectra, which differs between a three-dimensional box and the component-symmetric ODT line. Figure 10 and table 1 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial spectral centroid.

The structure of the table is the finding. Exact linear theory concentrates the strain-induced upwash anisotropy at the energy-containing scales and *removes* it from the small

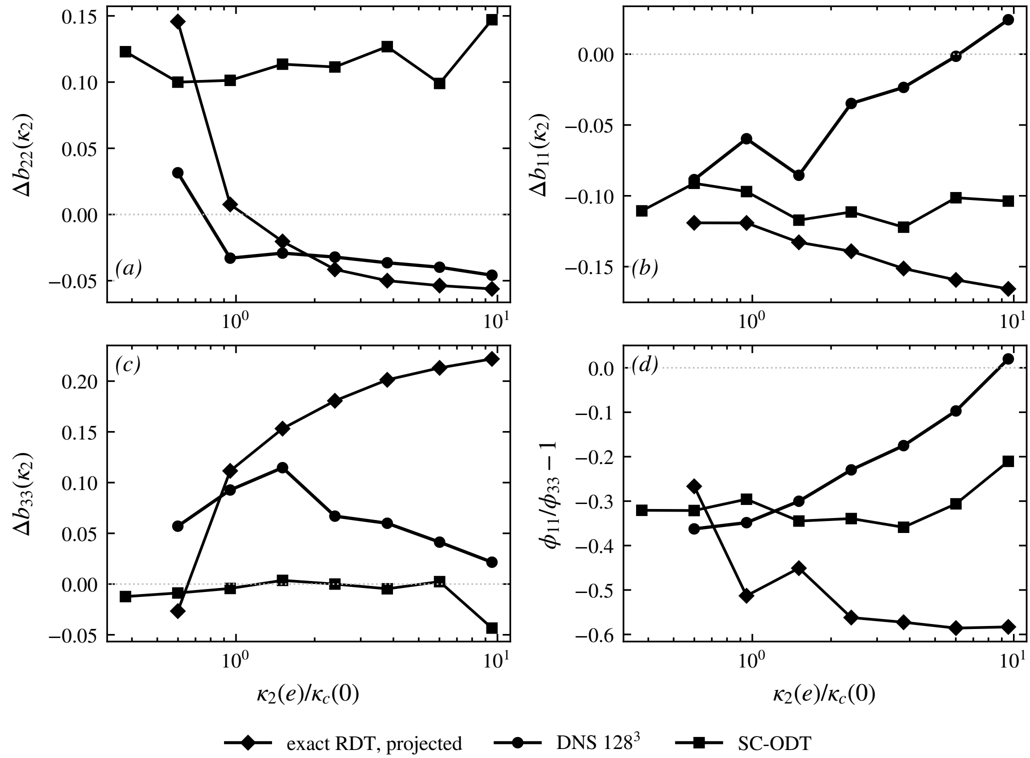

Figure 10. Three-way comparison at $e = 1$, $Sk_t/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra relative to each system's own unstrained state, resolved by component—(*a*) Δb_{22}, (*b*) Δb_{11}, (*c*) Δb_{33}, (*d*) the transverse splitting $\phi_{11}/\phi_{33} - 1$. SC-ODT (■, 1024 realisations) against exact projected RDT (♦) and DNS (128^3, •, four realisations). The SC-ODT curves are near-flat in every panel—the wavenumber-uniform signature—whereas exact RDT and the DNS vary with wavenumber.

scales, because modes aligned with the line are amplified least; the DNS shows the same shape, weakened at the large scales by the nonlinear return. SC-ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the signature, anticipated in § 4.2, of a wavenumber-uniform redistribution operator acting on a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks the closure at the moment level, § 3.4), so the deficiency is one of allocation across scales, not of amount. For the leading-edge application the operative consequence is that the high-wavenumber anisotropy of the upwash spectrum, which the Amiet response weights, has the wrong sign in SC-ODT; the closure element that must carry the correction—a wavenumber-dependent rapid operator in place of the uniform one—is exactly the spectral kernel formulation of § 5, whose exercise is thereby motivated by a measured deficit rather than by completeness.

4.6.2. *Distance from linear theory as a function of strain rapidity*

The three-way comparison resolves the anisotropy by wavenumber at one strain-rate parameter. A complementary scalar measure tracks the *whole* spectral shape across strain and strain rapidity: at each total strain the median ODT component line spectra are fitted, jointly in the upwash and transverse components, by the exactly distorted spectrum of linear theory—the closed-form Cauchy distortion of an isotropic von Kármán field, projected onto the line, with only the pre-distortion amplitude and energy wavenumber free. The root-mean-square log-residual of that two-parameter fit is a scale-resolved distance between

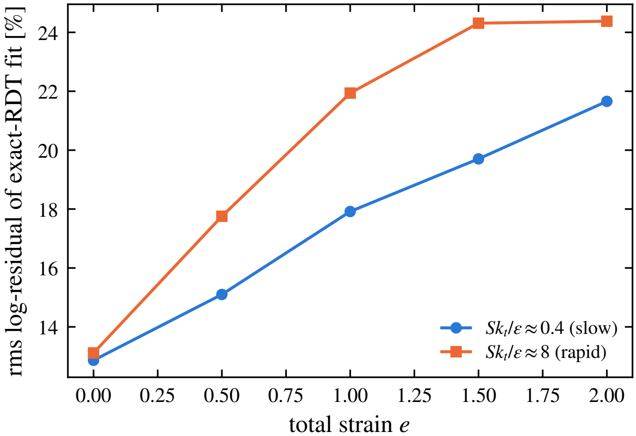

Figure 11. Scale-resolved distance between the strained SC-ODT line spectra and exact linear theory: rms log-residual of the two-parameter exact-RDT-distorted von Kármán fit, against total strain, for slow and rapid onset strain-rate parameters (medians over 1024 realisations; identical precursor state at $e = 0$).

the model's strained spectra and exact linear theory. Figure 11 shows it for precursor-conditioned ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 at onset (1024 realisations each, identical precursor state: the two fits coincide at $e = 0$ at thirteen per cent, the residual shape mismatch of a relaxed ODT line against the von Kármán form).

The distance grows with strain in both cases, but *faster at rapid strain*: to approximately 22 per cent at $e = 2$ for $Sk_t/\varepsilon \approx 0.4$ against approximately 24 per cent, saturating early, at $Sk_t/\varepsilon \approx 8$, with the separation established already by $e = 0.5$. If the model's rapid response were spectrally faithful, the rapid curve is the one that should approach zero: eddy events are subdominant there by construction. That the deviation is largest precisely where the eddies matter least localises the deficiency in the rapid kinematics themselves—the wavenumber-uniform operator and the rigid dilatation—and not in the eddy or relaxation statistics, in quantitative agreement with the three-way comparison. This measured validity envelope accompanies the model into any application: predictions inherit a spectral-shape uncertainty of order the plotted distance at the operating point's strain and rapidity.

5. Closure of the rapid pressure–strain term on a single line

The results of § 4 establish what SC-ODT produces and where it stands against exact linear theory and DNS. The rapid pressure–strain redistribution that drives that evolution was evaluated throughout by the algebraic moment closure of § 2 (Launder *et al.* 1975), applied component by component on the resolved line. That closure is a modelling input, and the redistribution it represents is, in its exact form, a *nonlocal, three-dimensional* object: the rapid pressure at a point is a volume integral over the entire fluctuating field, and the pressure–strain correlation is an integral over the full three-dimensional spectrum. What a single one-dimensional line of turbulence can say about such an object is the analytical question of this paper, and this section answers it in a form that is deliberately more modest, and more defensible, than a closure claim: we state exactly what obstructs a single-line representation, we derive sharp bounds on the rapid term from line data alone, we *measure* rather than assume the axisymmetry hypothesis under which the term collapses, and we identify—by measurement—which physical content the moment closure discards.

The section proceeds as follows. The exact term and the obstruction are stated first (§ 5.1–§ 5.2); the axisymmetric collapse follows, together with the statement of what it does *not* deliver (§ 5.3); the bounds that line data do deliver, with their falsifiers, come next (§ 5.4); the axisymmetry assumption is then measured directly in the strained-box DNS, including a control experiment that isolates its failure mode (§ 5.5); the component-ordering deficiency raised by the exact solution is measured and localised (§ 5.6); the reduction to the moment closure and the disjointness from the eddy redistribution close the argument (§ 5.7–§ 5.8).

5.1. *The exact rapid pressure–strain term*

The starting point is the Poisson equation (2.6) for the fluctuating pressure, whose source splits into a part linear in the fluctuations and the mean velocity gradient (the *rapid* part) and a part quadratic in the fluctuations (the *slow* part) (Pope 2000). The rapid part is linear in $\boldsymbol{u}'$ and proportional to the mean gradient, which for the present homogeneous plane strain is the constant tensor A_{kl}. Writing p'_{r} for the rapid pressure and inverting the Laplacian with the free-space Green's function $G(\boldsymbol{x}) = -1/4\pi|\boldsymbol{x}|$,

$$p'_{\mathrm{r}}(\boldsymbol{x}) = \frac{\rho}{2\pi} A_{kl} \int \frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|} \frac{\partial u'_l}{\partial x'_k}(\boldsymbol{x}')\,\mathrm{d}\boldsymbol{x}'. \tag{5.1}$$

The rapid pressure–strain correlation that redistributes energy among the Reynolds-stress components is $\mathit{\Pi}^{(\mathrm{r})}_{ij} = \overline{(p'_{\mathrm{r}}/\rho)\,(\partial u'_i/\partial x_j + \partial u'_j/\partial x_i)}$. Substituting (5.1) and using homogeneity to pass to the velocity spectrum tensor $\mathit{\Phi}_{mn}(\boldsymbol{\kappa})$ (the Fourier transform of $\overline{u'_m(\boldsymbol{x})\,u'_n(\boldsymbol{x}+\boldsymbol{r})}$), one obtains the standard exact result (Batchelor & Proudman 1954; Crow 1968; Hunt & Carruthers 1990)

$$\mathit{\Pi}^{(\mathrm{r})}_{ij} = A_{kl}\,M_{ijkl}, \qquad M_{ijkl} = \int_{\mathbb{R}^3} \left[\frac{\kappa_j\kappa_k}{\kappa^2}\,\mathit{\Phi}_{il}(\boldsymbol{\kappa}) + \frac{\kappa_i\kappa_k}{\kappa^2}\,\mathit{\Phi}_{jl}(\boldsymbol{\kappa}) \right] \mathrm{d}\boldsymbol{\kappa}, \tag{5.2}$$

where $\kappa = |\boldsymbol{\kappa}|$. Equations (2.6)–(5.2) are exact; no modelling has entered. The tensor M_{ijkl} is a linear functional of the *full three-dimensional* spectrum $\mathit{\Phi}_{mn}(\boldsymbol{\kappa})$ through the nonlocal projection operator $\kappa_a\kappa_b/\kappa^2$, which is the wavenumber-space image of the inverse Laplacian in (5.1). This nonlocality and three-dimensionality are the obstruction that the single-line model must confront.

5.2. *The obstruction: what an ODT line resolves*

ODT evolves a velocity vector $\boldsymbol{u}'(y)$ on a single line aligned with the resolved direction, here taken along the contraction axis of the strain, $y \equiv x_2$. From it the model has access to the one-dimensional spectra

$$\phi_{mn}(\kappa_2) = \int_{\mathbb{R}^2} \mathit{\Phi}_{mn}(\boldsymbol{\kappa})\,\mathrm{d}\kappa_1\,\mathrm{d}\kappa_3, \tag{5.3}$$

the one-dimensional spectra obtained by integrating $\mathit{\Phi}_{mn}$ over the two unresolved wavenumber components. The integrand of (5.2), by contrast, carries the *direction-resolved* weight $\kappa_a\kappa_b/\kappa^2$, which cannot be reconstructed from $\phi_{mn}(\kappa_2)$ alone: integrating out κ_1, κ_3 as in (5.3) destroys exactly the angular information that the projection operator acts on. The reduction $M_{ijkl} \to$ (functional of ϕ_{mn}) is therefore not available without an assumption about how the energy is distributed over the unresolved wavevector directions. This is the crux of the present question, and we make the required assumption explicit rather than implicit.

The obstruction just stated does not disappear under any symmetry assumption short of full isotropy. As § 5.3 shows, axisymmetry about the line reduces the unknown spectrum to two scalar functions of $(\kappa_2, \kappa_\perp)$—but the line data are two weighted $\kappa_\perp$-integrals of those functions per resolved wavenumber, so the map from what the line carries to what the rapid term requires retains an infinite-dimensional null space. Stating this null space exactly, and extracting what survives it, is the correct formulation of the single-line question.

5.3. *The closure assumption and the collapsed form*

The spectrum tensor of the incoming turbulence, before the rapid distortion acts, is axisymmetric about the resolved direction $\boldsymbol{e}_2$: $\Phi_{mn}(\boldsymbol{\kappa})$ depends on $\boldsymbol{\kappa}$ only through κ_2 and $\kappa_\perp = (\kappa_1^2 + \kappa_3^2)^{1/2}$, and its polarisation is invariant under rotation about $\boldsymbol{e}_2$.

This is a statement about the *turbulence*, not about the strain. It is justified for the present configuration because the inflow is grid turbulence, which is closely isotropic at the grid and is subsequently contracted along a single axis as it approaches the stagnation plane; isotropy is the special case of A1, and uniaxial contraction preserves axisymmetry about the contraction axis. The plane strain $A_{kl} = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ acting at the leading edge is itself *not* axisymmetric, so A1 is imposed on the state on which the rapid operator acts, consistent with the rapid-distortion ordering in which $\Pi^{(\mathrm{r})}$ is evaluated on the current (here, axisymmetric to leading order) field. The error incurred when the accumulated distortion has driven the spectrum away from axisymmetry is quantified in § 5.5.

Under A1 the angular integral in (5.2) can be carried out. Writing $\boldsymbol{\kappa} = (\kappa_\perp \cos\theta,\ \kappa_2,\ \kappa_\perp \sin\theta)$ and using that, by A1, Φ_{mn} is independent of θ, the azimuthal averages of the projection weights reduce to

$$\left\langle \frac{\kappa_1^2}{\kappa^2} \right\rangle_\theta = \left\langle \frac{\kappa_3^2}{\kappa^2} \right\rangle_\theta = \frac{1}{2}\,\frac{\kappa_\perp^2}{\kappa^2}, \qquad \left\langle \frac{\kappa_2^2}{\kappa^2} \right\rangle_\theta = \frac{\kappa_2^2}{\kappa^2}, \qquad \left\langle \frac{\kappa_1\kappa_3}{\kappa^2} \right\rangle_\theta = 0, \tag{5.4}$$

with $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and all weights mixing a resolved and an unresolved index (e.g. $\kappa_1\kappa_2/\kappa^2$) averaging to zero. Substituting (5.4) into (5.2) removes the azimuthal dependence and leaves a two-dimensional integral over $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. Defining the diagonal one-dimensional spectra along the line,

$$E_n^{\parallel}(\kappa_2) = \int_0^\infty \Phi_{nn}(\kappa_2, \kappa_\perp)\, 2\pi\kappa_\perp\, \mathrm{d}\kappa_\perp \quad \text{(no sum)}, \tag{5.5}$$

the rapid pressure–strain term reduces to a two-dimensional integral over the half-plane $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. The axisymmetric, solenoidal spectrum is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950); we denote them $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$, where a is the isotropic-like scalar (the sole survivor when the turbulence is isotropic) and c measures the axisymmetric departure from isotropy. Carrying out the azimuthal average (5.4) and specialising to the plane strain $A = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$, the diagonal rapid pressure–strain components are

$$\Pi_{nn}^{(\mathrm{r})} = \int_0^\infty\!\!\int_0^\infty g_n(x)\left[\, a(\kappa_2, \kappa_\perp),\ c(\kappa_2, \kappa_\perp) \,\right] 2\pi\kappa_\perp\, \mathrm{d}\kappa_\perp\, \mathrm{d}\kappa_2 \quad \text{(no sum on } n\text{)}, \tag{5.6}$$

with $x \equiv \kappa_2^2/\kappa^2$, $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and the kernels, linear in the two spectral scalars,

$$g_1(x) = a\left(\tfrac{1}{8} + \tfrac{3}{4}x - \tfrac{7}{8}x^2\right) + c\,\tfrac{7}{8}\,x\,(1-x)^2, \tag{5.7}$$

$$g_2(x) = -\tfrac{3}{2}\,x\left[\, a\,(1-x) + c\,(1-x)^2 \,\right], \tag{5.8}$$

$$g_3(x) = a\left(-\tfrac{1}{8} + \tfrac{3}{4}x - \tfrac{5}{8}x^2\right) + c\,\tfrac{5}{8}\,x\,(1-x)^2. \tag{5.9}$$

Equations (5.6)–(5.9) constitute the axisymmetric collapse: *under A1, the rapid pressure–strain term is a functional of the axisymmetric spectral scalars a, c on the resolved half-plane and the mean strain*, with no reference to the unresolved azimuthal structure beyond what A1 fixes. The kernels satisfy two exact constraints that verify the reduction. First, energy conservation: pressure–strain redistributes without changing the trace, and indeed $g_1(x)+g_2(x)+g_3(x) = 0$ identically for all x and for both scalars, so $\Pi^{(r)}_{11}+\Pi^{(r)}_{22}+\Pi^{(r)}_{33} = 0$. Second, the isotropic limit: setting $c = 0$ with a a function of κ alone, the integral of g_3 over the half-plane vanishes, so $\Pi^{(r)}_{33} = 0$—as it must, since the plane strain has no component along x_3 and the isotropic rapid term is proportional to S_{ij} (Crow 1968). Both constraints hold identically in the derived kernels, confirming the tensor reduction.

The status of (5.6)–(5.9) must be stated precisely, because it is here that the closure question is decided. Under A1 the rapid term is a linear functional of the two axisymmetric scalars $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$ on the resolved half-plane—a *two-dimensional* representation, not yet a single-line closure, since the line determines only the $\kappa_\perp$-integrals (5.5) of fixed combinations of a and c and not the functions themselves. Equations (5.6)–(5.9) are therefore the *collapsed form* on which any single-line statement must be built; the single-line content itself—what the line data pin down, and how sharply—is established next.

5.4. *What line data determine: sharp bounds on the rapid term*

Under A1 the solenoidal spectrum tensor takes the two-scalar form $\Phi_{mn} = a\,P_{mn} + c\,\xi_m\xi_n$, with P_{mn} the solenoidal projector and ξ_m the projected axis direction; realisability is the pair of pointwise conditions $a \geq 0$ and $a + (1-x)\,c \geq 0$ with $x = \kappa_2^2/\kappa^2$. The line carries $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ and $\phi_{22}(\kappa_2)$: per resolved wavenumber, two weighted $\kappa_\perp$-integrals of the two unknown functions. The rapid term (5.6) is linear in (a, c), so its extreme values over everything consistent with the line data, solenoidality and realisability are obtained from a family of linear programs, one per κ_2, yielding sharp bounds $[\Pi^-_{nn}, \Pi^+_{nn}]$ and a kinematic indeterminacy $\mathcal{I}_n = (\Pi^+_{nn} - \Pi^-_{nn})/2k_t S$ that quantifies exactly how much the null space costs. The forward map from (a, c) to the line spectra and the derivation of the bound kernels are given in Appendix A.

Calibrated on an isotropic von Kármán spectrum (with a verification suite pinning tracelessness, the isotropic angular averages $+\frac{1}{5} : -\frac{1}{5} : 0$, the longitudinal–transverse relation, and the exact $\frac{4}{5}k_t S_{ij}$), the indeterminacy is

constraint level	$\mathcal{I}_{11}$	$\mathcal{I}_{22}$	$\mathcal{I}_{33}$
line data + solenoidality + realisability	0.77	1.28	0.86
+ log-Lipschitz slope cap ($\lambda = 4$)		< 5% narrower	
+ polarisation cap $\lvert c\rvert \leq \gamma a$, $\gamma = 0$	0.34	0.58	0.24

Two features of this table matter. Smoothness assumptions buy almost nothing: the null space is not a roughness artefact. The polarisation cap is the lever—but § 5.5 shows that under strain the effective polarisation reaches order one, so the honest bound in the strained regime is the uncapped one.

The representation also supplies two *free falsifiers* that the line itself reports: under A1, $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ at every wavenumber (T1) and the velocity cross-spectra vanish (T2). Any measured transverse splitting is a wavenumber-resolved violation of A1; rather than assuming the hypothesis, the line monitors it.

	$Sk_t/\varepsilon = 0.8$				$Sk_t/\varepsilon = 16$			
e	b_{22}	γ_{eff}	m=2	$\frac{\phi_{11}}{\phi_{33}}-1$	b_{22}	γ_{eff}	m=2	$\frac{\phi_{11}}{\phi_{33}}-1$
0	0.003	0.10	0.133	+0.06	0.003	0.10	0.133	+0.06
0.5	0.064	0.79	0.166	−0.20	0.070	1.28	0.247	−0.21
1	0.125	1.44	0.189	−0.32	0.124	4.22	0.396	−0.46

Table 2. A1 measured under plane strain (128^3 DNS, four realisations): upwash anisotropy, effective polarisation, azimuthal $m = 2$ residue of Φ_{22} (isotropic floor 0.133), and the on-line falsifier T1.

5.5. *The axisymmetry hypothesis, measured*

A1 is exact for the isotropic inflow and is degraded by the plane strain itself. Rather than estimating this error perturbatively, we measure it in the strained-box DNS of § 4.6, where the full spectrum is available: the exact azimuthal decomposition of Φ_{22} yields the true scalars (a, c), the discarded $m = 2$ azimuthal residue (the direct A1-violation measure, with isotropic sampling floor 0.133 in this box), and the effective polarisation $\gamma_{\mathrm{eff}} = |c|/a$.

Three conclusions follow (table 2). First, the A1 error is *not* perturbatively small in the accumulated anisotropy: the residue grows from its floor to 0.19 at $Sk_t/\varepsilon = 0.8$ and to 0.40 at 16 by $e = 1$—larger at rapid strain, because at moderate strain rate the nonlinear dynamics restore axisymmetry as fast as the strain destroys it. An $O(b^2)$ error estimate, which the symmetric structure of the discarded harmonic suggests, is superseded by this measurement. Second, the polarisation cap that narrows the isotropic bounds is illegitimate under strain: γ_{eff} passes unity by $e \simeq 0.5$. Third, correspondingly, the uncapped bound brackets the exact rapid term (evaluated by direct integration in the DNS) up to $e \approx 0.25$–0.5 and fails beyond, and the failure is A1's, not the estimator's.

The decisive evidence for that attribution is a control experiment: the same box under *axisymmetric contraction* with matched A_{22}, for which A1 is exact by symmetry. There the $m = 2$ residue stays at the sampling floor and the transverse splitting at zero at every strain up to $e = 1$ even at $Sk_t/\varepsilon = 16$, while under plane strain both grow (figure 12). The single-line closure is therefore *exact for axisymmetric distortion*; its plane-strain error is a strain-geometry property, now quantified, rather than a defect of the line representation. For the applications this maps cleanly: a rounded, three-dimensional stagnation region imposes an axisymmetric strain (closure exact); a two-dimensional leading edge imposes plane strain (closure carries the measured correction). The acoustically relevant amplification $b_{22}(e)$, finally, is indistinguishable between the two geometries (0.124 against 0.117 at $e = 1$, $Sk_t/\varepsilon = 16$): the quantity the leading-edge response consumes is robust to precisely the geometry that A1 is sensitive to.

5.6. *Component ordering under plane strain*

The exact solution of the rapid problem under plane strain has the neutral spanwise component eventually overtaking the upwash; algebraic closures predict the opposite. The exact rapid term evaluated in the DNS settles the matter: at $e = 1$, $\Pi^{(\mathrm{r})}_{33}/k_t S = 0.14$ at $Sk_t/\varepsilon = 0.8$ and 0.27 at 16, exceeding $\Pi^{(\mathrm{r})}_{11} = 0.23$ in the rapid case—the overtaking is real and sets in at rapid strain. LRR-QI, whose spanwise component is carried by its b-dependent term, matches at $Sk_t/\varepsilon = 0.8$ ($\Pi^{(\mathrm{r})}_{33} = 0.143\, k_t S$) but delivers half the exact value at 16; IP gives zero identically. The deficiency thus lives in the moment closure's b-term, not in the line model: the ODT moment-level b_{ij} follows the closure to $\sim 2\%$, and

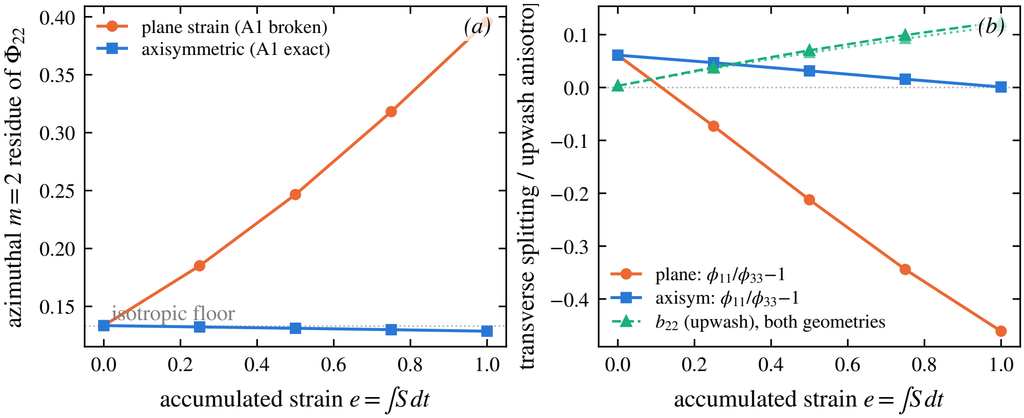

Figure 12. The A1 control experiment at $Sk_t/\varepsilon = 16$ (128^3, four realisations): axisymmetric contraction (A1 exact by symmetry) against plane strain at matched A_{22}. Left: azimuthal $m = 2$ residue of Φ_{22}. Right: transverse splitting and b_{22}.

its b_{33} stays within 0.006 of zero in every run. In the spectral representation the ordering is carried by the c-scalar through the kernels (5.7)–(5.9); the moment reduction discards it. This answers, by measurement, where the known component-ordering error enters and what it would take to remove it.

5.7. *Reduction to the moment closure*

The closed form (5.6)–(5.9) operates on the resolved spectrum, whereas the distortion results of § 4 were obtained with the algebraic moment closure of § 2.4 (Launder *et al.* 1975). For the two to be consistent, the spectral kernel must reduce to that moment closure in the limit where the moment closure is exact—isotropic turbulence. We show that it does, which both justifies the closure used in the simulations and fixes the prefactor in an explicit convention.

Figure 13 plots the kernels g_1, g_2, g_3 of (5.7)–(5.9) against the angular variable $x = \kappa_2^2/\kappa^2$. Two features are exact and hold for both spectral scalars. The kernels sum to zero at every x,

$$g_1(x) + g_2(x) + g_3(x) = 0 \qquad \forall x, \tag{5.10}$$

so the rapid term conserves turbulent kinetic energy pointwise in wavenumber, not merely in the integral; and each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection $\kappa_a\kappa_b/\kappa^2$ degenerates. The signs are physically ordered: $g_1 > 0$ (energy fed to the streamwise component E_1), $g_2 < 0$ (energy removed from the line/upwash component E_2) over the bulk of the angular range, with g_3 changing sign, consistent with the plane strain stretching x_1 and compressing x_2.

For *isotropic* turbulence the anisotropy scalar vanishes ($c = 0$) and the isotropic-like scalar a depends only on κ. Averaging the kernels over solid angle—equivalently, integrating $g_n(x)$ against the isotropic distribution of x on the sphere—gives

$$\langle g_1\rangle : \langle g_2\rangle : \langle g_3\rangle = +\tfrac{1}{5} : -\tfrac{1}{5} : 0, \tag{5.11}$$

in the ratio $+1 : -1 : 0$. This is exactly the structure of the plane strain $S_{ij} = \mathrm{diag}(\tfrac{1}{2}, -\tfrac{1}{2}, 0)$: the rapid term is proportional to S_{ij}, with $\Pi_{33}^{(\mathrm{r})} = 0$ because the strain has no component along x_3. The spectral kernel therefore collapses, under isotropy, to the isotropic-production

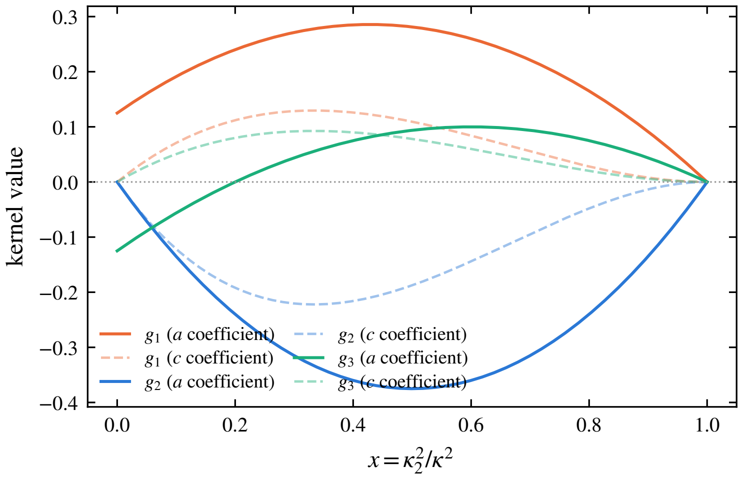

Figure 13. Rapid pressure–strain closure kernels g_1, g_2, g_3 (eqs. (5.7)–(5.9)) against the angular variable $x = \kappa_2^2/\kappa^2$. Solid curves: coefficient of the isotropic-like spectral scalar a; faint dashed curves: coefficient of the anisotropy scalar c. The kernels sum to zero at every x (energy conservation, eq. (5.10)) and vanish at $x = 1$. Averaged over an isotropic spectrum they give the ratio $+\frac{1}{5} : -\frac{1}{5} : 0$ (eq. (5.11)), recovering the isotropisation-of-production form $\Pi_{ij}^{(\mathrm{r})} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) that underlies the moment closure used in § 4.

form of the moment closure,

$$\Pi_{ij}^{(\mathrm{r})} = \tfrac{4}{5}\, k_t\, S_{ij} \qquad \text{(isotropic limit)}, \tag{5.12}$$

the result of Crow (1968) that underlies the isotropisation-of-production (IP) rapid term in the closure of Launder *et al.* (1975). We note a convention: with the symmetrised definition of the pressure–strain correlation adopted here and the factor of two carried in the Poisson source (2.6), the kernels (5.7)–(5.9) yield the coefficient $\frac{2}{5} k_t$ for the 11-component directly (since $S_{11} = \frac{1}{2}$), equivalent to the standard $\frac{4}{5} k_t S_{ij}$ of (5.12); the ratio (5.11) and the tracelessness (5.10) are independent of this convention.

The consequence is that the algebraic closure used to obtain the distortion of § 4 is not an independent assumption but the isotropic-limit reduction of the line-consistent spectral term derived here. The spectral form (5.6) carries, in addition, the angular (x-dependent) and anisotropy (c-scalar) information that the moment closure discards—and both discards are now measured: the wavenumber dependence of the linear response in the three-way comparison of § 4.6.1, and the component ordering in § 5.6. The moment closure and the collapsed form thus agree to leading order, and the terms in which they depart are exactly the measured deficiencies of § 4.

5.8. *Consistency with the eddy redistribution: avoiding double counting*

A closed rapid term is necessary but not sufficient: it must not duplicate redistribution already effected by the eddy events. The pressure–strain correlation decomposes canonically into rapid and slow parts, $\Pi_{ij} = \Pi_{ij}^{(\mathrm{r})} + \Pi_{ij}^{(\mathrm{s})}$, corresponding to the two source terms in (2.6). The slow part $\Pi_{ij}^{(\mathrm{s})}$ is the return-to-isotropy mechanism, quadratic in the fluctuations and independent of the mean gradient; the eddy events in ODT, which act on the fluctuating field without reference to A_{kl}, are the model's representation of exactly this slow redistribution. The rapid part $\Pi_{ij}^{(\mathrm{r})}$, proportional to A_{kl} and linear in the fluctuations, is by construction absent from the eddy mechanism, which carries no dependence on the mean strain. The two therefore occupy disjoint terms of the canonical decomposition, and adding (5.6) as an explicit mean-strain-driven source alongside the eddies does not double-count, provided the eddy events are calibrated to reproduce the slow return-to-isotropy in the absence of mean strain—a condition that the no-strain baseline of § 4.3 verifies directly, since there the centroid relaxes under the eddies alone with no spurious rapid contribution. This argument establishes consistency at the level of the decomposition; the no-strain baseline of § 4.3 provides the direct numerical confirmation, the centroid there relaxing under the eddy mechanism alone.

The disjointness argument is now also a measured statement, from both sides. With the strain switched off the eddy events generate no spurious strain-like signal (the nominal-turbulence null of Appendix B). Conversely, in the rapid limit ($Sk_t/\varepsilon = 16$, under one eddy per line in $e \leq 1$) every kernel-energy allocation rule tested—the isotropic kernel, a strain-biased generalisation, and discrete eddy types—reproduces the exact rapid-distortion evolution of $b_{22}(e)$ to ± 0.003: the rapid term cannot be reached from inside the eddy events, and the continuous operator carries it alone. Rapid and slow redistribution occupy disjoint terms by construction, and now by measurement.

5.9. *Summary: what the single-line treatment establishes*

1. The rapid term is exactly $A_{kl}M_{ijkl}$; its single-line representation is obstructed by an azimuthal null space that is stated exactly and survives the axisymmetric collapse at the level of the two spectral scalars.
2. Under A1 the term collapses to the kernel form (5.6)–(5.9); A1 is exact for axisymmetric distortion and is *measured* under plane strain, where its violation grows with accumulated strain and with strain rapidity (table 2).
3. Independently of A1's validity, line data bound the rapid term sharply; the bound is honest—and verified to bracket the exact term—up to $e \approx 0.25$–0.5 under plane strain, and at all computed strains under axisymmetric distortion.
4. The moment closure used in the simulations is the isotropic reduction of the collapsed form (verified, including the exact $\frac{4}{5}k_t S_{ij}$); what it discards is the wavenumber dependence of the linear response and the component ordering, both measured in § 4 and § 5.6.
5. Rapid and slow redistribution are disjoint: verified by the strain-off null and by the rapid-limit invariance of every kernel-energy allocation rule.
6. The acoustically relevant upwash amplification $b_{22}(e)$ is linear to $e = 1$ and robust to the strain geometry that controls A1's validity.

6. Measured limits, and scale-local relaxation

The envelope of § 4.6.2 localises the model's spectral deficiency in its wavenumber-uniform rapid kinematics: the strain-induced anisotropy is allocated uniformly across scales where linear theory and DNS concentrate it at the energy-containing range and remove it from the fine scales. This section asks the constructive question: which mechanism available to the model can supply the missing scale dependence? We answer it experimentally, by a scale-resolved diagnostic and a controlled family of interventions in the eddy mechanism, each tested at 1024 realisations with seeds paired to a common baseline, and each either falsified or confirmed by the same measurement. The outcome is a single measured statement: what is missing is *scale-local relaxation*, and supplying it—at the scales of the imbalance—is both necessary and, at moderate strain, sufficient to move the fine-scale anisotropy.

6.1. *A transmission diagnostic*

The configuration is a scale-diagnostic variant of the strained line: unit strain rate (so $e = t$), an initial spectrum peaked at a wavenumber κ_p resolved by more than a decade of smaller scales, and two measurement bands, $\kappa_2 \in [0.6, 2]\,\kappa_p$ (the energy-containing band) and $[6, 16]\,\kappa_p$ (roughly two triplet-map generations below it, since each map compresses by a factor of three). On each band the component imbalance is $\mathcal{A}_b = E_2^{\parallel}(b)/\frac{1}{2}[E_1^{\parallel}(b)+E_3^{\parallel}(b)]$, the band-integrated upwash energy over the transverse mean; all statistics are medians over realisations with bootstrap confidence intervals, for the intermittency reason given in § 4.6. The model's eddy and kernel mechanisms are component-symmetric, so $\mathcal{A} = 1$ is their fixed point; the strain drives $\mathcal{A}$ up at the scales where it deposits anisotropy, and the ratio $\mathcal{A}_{\mathrm{hi}}/\mathcal{A}_{\mathrm{lo}}$ measures how much of the large-scale imbalance survives transmission down the cascade.

The baseline (supplementary material, figure S3, black; also the black curve of figure 14) exhibits the deficiency of § 4.6.1 in transmission form: $\mathcal{A}_{\mathrm{hi}}/\mathcal{A}_{\mathrm{lo}}$ stays near unity out to $e = 3$ (1.02, 0.95, 0.89 at $e = 1, 2, 3$)—the anisotropy injected at the energy-containing scales crosses two map generations essentially undiminished—and drops only at the deepest strain. The mechanism is direct: the high band is populated by the maps of *large* eddies (one generation takes κ_p to $3\kappa_p$, a second to $9\kappa_p$, inside the band), so fine-scale anisotropy is delivered by large-eddy events, not by a chain of small ones.

6.2. *Interventions: what fails and what works*

The complete intervention family—each case's knob, measured fine-scale effect and reading, with the four-panel diagnostic figure for the eddy-locked variants—is tabulated in the supplementary material (§ S2, table S1 and figure S3); the ledger it establishes is set out here, with the two decisive figures (the relaxation clock and the five-way benchmark) retained below. Two families fail for instructive reasons. Acceptance gating—rejecting accepted eddies whose post-event state remains anisotropic, at thresholds spanning 45 to 98 per cent rejection—is scale-blind: at mild thresholds the surviving eddy population reproduces the baseline anisotropy budget in every band (the gate self-selects eddies whose kernels were already going to do the relaxing, and the unrelaxed field keeps generating candidates), while at the harshest threshold the model merely drifts towards its eddy-free limit. Restricting the gate to sub-band-scale eddies sharpens the negative result into the mechanism statement above: removing up to 97 per cent of small-eddy events leaves both the fine-scale anisotropy *and* the fine-scale energy unchanged, so no acceptance-gating scheme can control transmission at this Reynolds number. Reweighting the triplet-map images (a middle image of relative size $\frac{2}{3}$, which halves the compression of the map's dominant central channel) moves one-point and energy-containing statistics—$\mathcal{A}_{\mathrm{lo}}$ falls significantly and the upwash energy fraction with it—but not the fine scales: with the image fractions summing to one, slowing the middle channel necessarily speeds the outer ones, and one outer image maps the energy peak directly into the measurement band.

What works is relaxation applied *at the scale of the imbalance*. After each accepted eddy's map and kernels, the same energy-equalising kernel procedure (§ 2) is applied to each of the three map images—the sub-interval's own triplet-map displacement shape as the kernel, coefficients from the sub-interval's own integrals, so that each sub-application conserves the momentum components and the total kinetic energy exactly, with no map applied and no random numbers drawn—and then recursively to 9 and 27 sub-regions. One level ($\ell/3$ for an eddy of size ℓ) is null: for the large eddies that feed the band, $\ell/3$ still sits above it. Two levels produce the first statistically significant fine-scale isotropisation of the family ($\mathcal{A}_{\mathrm{hi}}$ from 1.43 to 1.27 at $e = 1$, both bands' paired contrasts individually significant), and three levels extend the effect to deep strain: the levels that work, $\ell/9$ and $\ell/27$, are exactly the first that reach the measured band. The bookkeeping is clean throughout—band energies move by a few per cent, the upwash energy fraction and the intermittency tails are unchanged—and the pre-strain fine-scale floor moves towards component equality immediately ($\mathcal{A}_{\mathrm{hi}}$ from 0.83 to 0.95 at $e = 0$).

Four further variants separate the candidate explanations for the one remaining limitation, the fading of the gains beyond $e \approx 3$. Tying the recursion depth to the eddy size (subdividing while the sub-interval remains above a near-dissipative floor, so that large eddies do proportionally more sub-scale work) gives the gains of depth with no reversal at the deepest strain: both bands significant at $e = 1$ and the effect retaining its sign through $e = 3$. Iterating the two-level pass three times within each event answers the amplitude hypothesis in the negative: it is a stronger early isotropiser but overshoots at deep strain. Delocalising the relaxation—after each eddy, applying the kernel isotropisation in N randomly placed intervals of the same size, overlaps allowed—tests whether the *location* of the events is the limiter: $N = 1$ is null, and $N = 3$ is significant in both bands at $e = 1$ but acts mainly at the energy-containing scales, since the sampled intervals inherit the eddy-size distribution and full-interval isotropisation acts at that scale. Stacking the two ideas—the adaptive hierarchy inside each sampled interval—is the strongest early isotropiser of the family ($\mathcal{A}_{\mathrm{hi}}$ from 1.43 to 1.26 at $e = 1$) and delivers the only significant deep-strain effect, in the energy-containing band at $e = 3.9$; yet at the fine scales and deep strain it overshoots the depth-only variant significantly. The ledger is then complete for every scheme locked to the eddy clock: depth sets *where* the relaxation acts and is necessary; dosage and delocalisation raise the early gains in proportion to the total relaxation delivered; and no eddy-locked combination—more depth, more dosage, more events, other locations—prevents the fine-scale fade beyond $e \approx 3$. The remaining suspect is the clock itself: relaxation acts at the eddy rate while the strain resupplies the imbalance continuously. Whether that is fundamental or merely structural is decided by the final intervention.

6.3. *Decoupling the relaxation clock*

The one axis the eddy-locked family cannot test is *when* relaxation acts. We therefore give it its own clock: an independent Poisson stream of relaxation-only events at a prescribed rate, each event an interval drawn from the model's eddy-size distribution, placed uniformly on the domain, and relaxed by the same conservative kernel pass with the adaptive-depth hierarchy inside it. Because each event acts on an interval, momentum and kinetic energy are conserved exactly per event; because the stream has its own rate, its pacing is decoupled from the eddy statistics entirely. The rate is matched to the baseline's mean eddy rate, so the comparison against the eddy-locked variants is dose-matched by construction.

The result (figure 14) decides the question. With the clock, the fine-scale imbalance is reduced significantly at *every* strain and the effect grows rather than fades—the paired contrasts reach -0.30 (low band) and remain significant in the high band at $e = 3.9$—because a constant-rate stream keeps pace precisely where the eddy rate could not: event-locked timing, not any per-event property, was the deep-strain limiter. But the uncapped stream is scale-blind and overcorrects: it also suppresses the one-point anisotropy that the strain should produce, pulling $u_2^2/2k_t$ from 0.58 to 0.41 at $e = 3.9$ and thereby breaking the

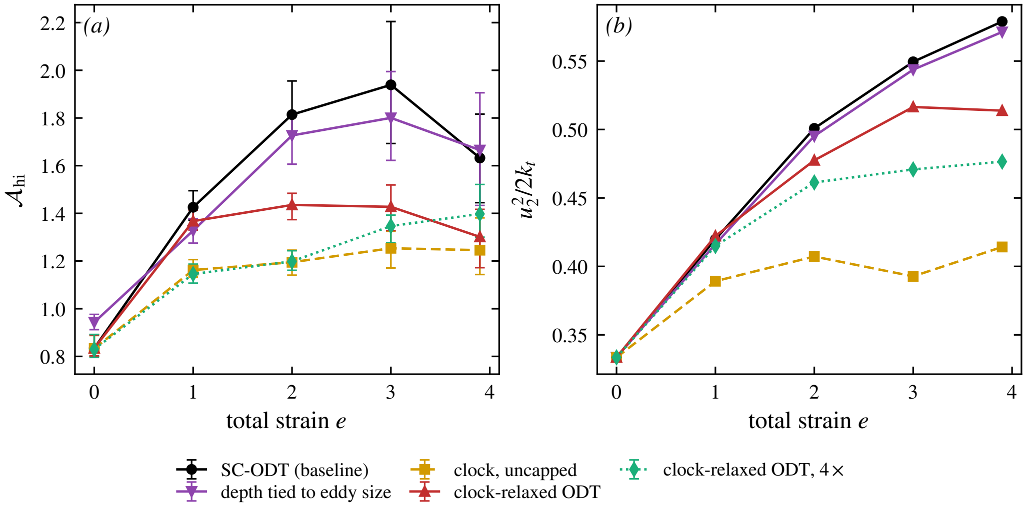

Figure 14. The independent relaxation clock against the eddy-locked family. (*a*) Fine-scale-band imbalance $\mathcal{A}_{\rm hi}$ against total strain: the eddy-locked variants fade beyond $e \approx 3$; the clock holds at every strain. (*b*) The price: upwash energy fraction—the uncapped clock suppresses the physical one-point anisotropy, the size cap restores it. Medians over 1024 paired-seed realisations; the clock rate is matched to the baseline eddy rate.

moment-level behaviour validated in § 3.4. Restricting the stream to sub-band interval sizes ($\ell \leq \ell^*$, the scale separating the two measurement bands) resolves the tension: the size-capped clock holds the fine scales at every strain with no fade—$\mathcal{A}_{\rm hi}$ at 1.30–1.44 against the baseline's 1.63–1.94, contrasts significant throughout and strongest at $e = 3$—while the one-point trajectory is essentially untouched in the moderate-strain regime ($u_2^2/2k_t$ within 0.005 of the baseline at $e = 1$) and only mildly reduced at the deepest strain. The rate is a clean dial: four times the rate holds the fine scales harder at a proportionate one-point cost.

6.4. *The measured missing ingredient*

The arc closes the loop opened by the three-way comparison. The deficiency measured there—anisotropy allocated uniformly over wavenumber—is confirmed here in transmission form, is unfixable by gating or map reweighting, and is cured by exactly two ingredients, each isolated by a falsified alternative: relaxation applied *at the scale of the imbalance* (depth, not dosage or location) and relaxation applied *on its own clock* (the size-capped stream, which alone survives deep strain). One limit remains, and it is structural rather than curable by any amount of relaxation: an isotropising mechanism is bounded at component equality, whereas exact linear theory and the DNS carry the fine-scale upwash *below* the transverse mean. Crossing that boundary requires the rapid operator itself to discriminate by wavenumber, and the formulation already contains its natural form: the spectral kernel (5.6)–(5.9) is precisely a wavenumber- and angle-dependent rapid operator, whose discretisation in place of the wavenumber-uniform moment closure of § 2 is the identified next step.

The head-to-head that summarises this part is figure 15: the strain-induced upwash anisotropy per wavenumber band—the observable on which § 4.6.1 established the benchmark—for exact linear theory, the DNS, the strain-coupled model, and the model under each of the two mechanism forms (adaptive depth; the size-capped clock), at three strain-rate parameters, from same-binary ensembles under the identical precursor protocol. Four

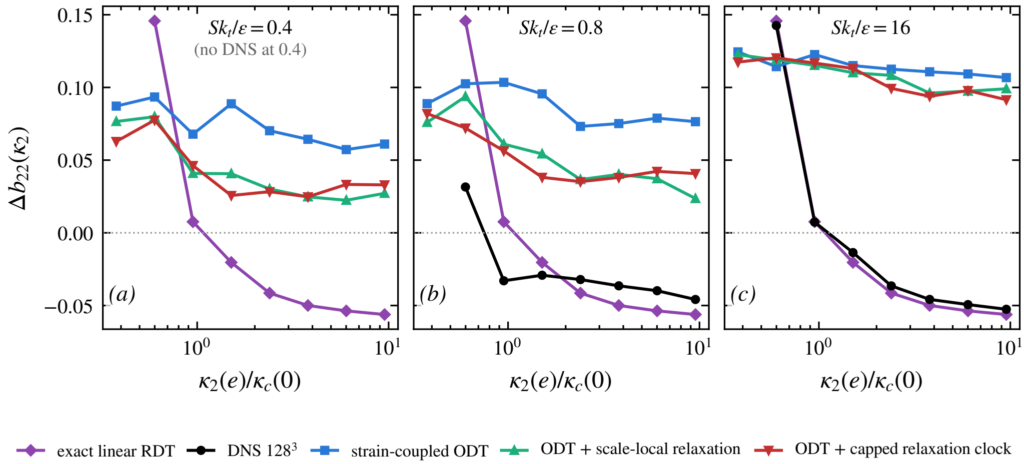

Figure 15. The five-way comparison on the benchmark observable of § 4.6.1: strain-induced upwash anisotropy per wavenumber band at $e = 1$, for exact linear RDT, DNS, the strain-coupled model, and the model with each mechanism form (adaptive depth; size-capped relaxation clock), at $Sk_t/\varepsilon = 0.4$, 0.8 and 16 (ODT: 1024 realisations per case, same binary and precursor protocol; DNS and companions as in § 4.6; no DNS exists at 0.4, where exact RDT is the reference—at fixed total strain it is independent of the strain rate). At rapid strain the DNS realises the exact-RDT allocation; the mechanism moves the model towards the DNS increasingly as the rapidity falls.

statements can be read directly from the shared axes. First, at $Sk_t/\varepsilon = 16$ the DNS lies on the exact-RDT curve: the rapid limit is realised in the benchmark itself, and the deficiency of the baseline model—its uniform allocation of a correct integrated anisotropy—is starkest precisely where the eddies matter least. Second, at moderate rapidity the mechanism tilts the model's allocation towards the DNS—the anisotropy now decreases with wavenumber, the fine-band excess roughly halved (at $Sk_t/\varepsilon = 0.8$: from +0.076 against the DNS value −0.046 to +0.024)—and the improvement grows as the rapidity falls, the trend the application regime sits on. Third, within $e \leq 1$ the two mechanism forms are equivalent on these axes; the clock's advantage is the deep-strain hold of figure 14, which this protocol—bounded at $e = 1$ by the DNS reference—cannot display. Fourth, at $Sk_t/\varepsilon = 16$ neither form moves the allocation: rapid strain limits the physical time available, and no relaxation scheme manufactures more of it. The remaining, sign-level gap belongs to the wavenumber-dependent operator identified above.

7. Consequence for leading-edge noise

The application that motivates the model can now be served with its output. In the leading-edge-noise theory of Amiet (1975) the far-field pressure spectral density of a large-aspect-ratio aerofoil factorises as a product of the flow and observer geometry, the aerofoil response $|\mathcal{L}|^2(K_x)$, the upwash wavenumber spectrum $\varPhi_{ww}(K_x, 0)$ and the spanwise correlation length $\ell_y(K_x) = \pi\,\varPhi_{ww}(K_x, 0)/\int \varPhi_{ww}(K_x, K_y)\,\mathrm{d}K_y$, with K_x the streamwise and K_y the spanwise wavenumber. For a *change of inflow spectrum* at fixed aerofoil, flow and observer, everything except the turbulence kernel

$$\mathcal{K}(K_x) \;=\; \pi\,\frac{\varPhi_{ww}(K_x, 0)^2}{\int \varPhi_{ww}(K_x, K_y)\,\mathrm{d}K_y} \tag{7.1}$$

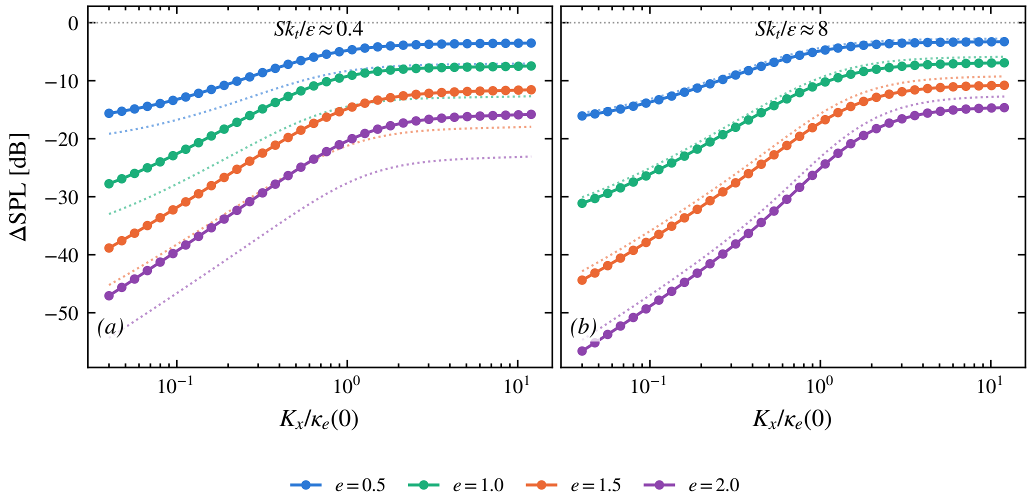

Figure 16. Change of the Amiet leading-edge-noise kernel when the frozen inflow spectrum is replaced by the computed distorted one, against reduced streamwise wavenumber, at total strains $e = 0.5$–2: (*a*) $Sk_t/\varepsilon \approx 0.4$, (*b*) ≈ 8. Solid: spectral reallocation at matched upwash variance (shape only); dotted: total, including the measured variance change (-3.5 to -7.3 dB at the slow rate, where decay dominates the transit; $+0.5$ to $+1.9$ dB at the rapid rate, where the upwash amplification survives). Geometry, aerofoil response and observer position cancel identically in this ratio.

cancels exactly, so the change of radiated level is $\varDelta\mathrm{SPL}(K_x) = 10\log_{10}[\mathcal{K}^{\mathrm{distorted}}/\mathcal{K}^{\mathrm{frozen}}]$, independent of the aerofoil geometry and response. This is the sharpest question the distortion model can be asked without importing an acoustic solver: by how much does the prediction change when the inflow spectrum is the *computed, distorted* one instead of the frozen upstream spectrum that standard practice prescribes?

The distorted input is taken from the model's own strained ensembles in the representation already fitted in § 4.6.2: the exact-RDT-distorted von Kármán family, two parameters (A_0, κ_e) per strain, fitted to the median line spectra of the 1024-realisation ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8. The line data pin the resolved-wavenumber structure; the transverse structure that (7.1) requires is supplied by the representation—this is § 5's statement about line knowledge, operating in the application—and the representation's rms residual, 13–24% over the strain range (the distance envelope of figure 11), is the corresponding uncertainty carried by the result. The two-dimensional spectrum follows by direct integration of the distorted components, $\varPhi_{ww}(K_x, K_y; e) = A_0 \int \varPhi_{22}^{\mathrm{RDT}}(K_x, \kappa_2, K_y; e)\,\mathrm{d}\kappa_2$, with the machinery verified against the wavevector-ensemble integrator of § 3.2 (upwash-variance growth reproduced to under 1%). The frozen baseline is the same inflow's relaxed $e = 0$ state carried undistorted. The level and shape contributions are separated: the kernel scales linearly with the upwash variance, so dividing it out isolates the spectral reallocation, and the variance ratio itself is taken from the measured line spectra rather than from the fit.

Figure 16 shows the result, and its sign and structure are the finding. At matched upwash variance the distorted spectrum is a substantially *weaker* leading-edge source than the frozen one at every resolved reduced frequency: ≈ -5 dB at $K_x \approx \kappa_e(0)$ for $e = 0.5$, deepening to -20 dB and beyond by $e = 2$, and increasingly severe towards low K_x. The mechanism is the orientation physics of § 5 acting on the noise kernel: (7.1) weights the spanwise-coherent upwash content $\varPhi_{ww}(K_x, 0)$, which lives on wavevectors aligned with the line—exactly the orientations whose upwash is smallest by solenoidality and whose pre-images retreat into the vanishing low-wavenumber tail of the initial spectrum as the

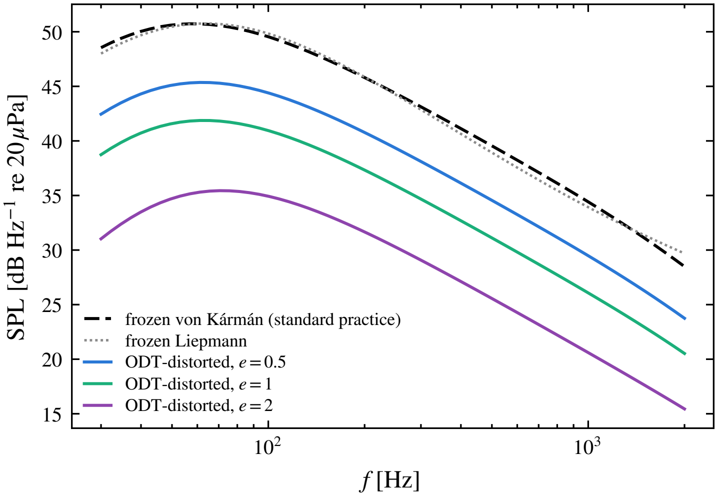

Figure 17. The assembled absolute prediction at the reference configuration of § 4.4: far-field sound pressure level from the compact lift-dipole form of Amiet (1975) with the exact Sears response, for a flat plate of chord 0.39 m ($Re_c \approx 5 \times 10^5$) and span 0.8 m at $U_\infty = 20\,\mathrm{m\,s^{-1}}$, $u'/U_\infty = 6\%$, $\mathcal{L} = 55$ mm, observer 1 m overhead. Dashed: the frozen von Kármán inflow of standard practice (the $e = 0$ member of the fitted representation, which is exactly the isotropic von Kármán spectrum); grey dotted: the frozen Liepmann spectrum at the same $(u', \mathcal{L})$, an independent analytic cross-check of the physical-unit bridge. Solid: the same chain with the computed distorted inflow at total strains $e = 0.5$, 1 and 2 ($Sk_t/\varepsilon \approx 0.4$), level and shape both included.

distortion proceeds. The distortion, in short, drains the very content the aerofoil responds to. The total change adds the variance term: at the rapid rate the surviving upwash amplification offsets a small part of the reallocation; at the slow rate viscous decay over the longer transit deepens it.

Figure 17 places the kernel result in the quantity practice actually reports: the absolute far-field spectrum of the reference configuration of § 4.4, assembled from the compact lift-dipole form of Amiet's theory with the exact Sears response. The spectra enter in physical units through two matching conditions only—the integral scale fixes the length unit through the von Kármán relation $\kappa_e = 0.7468/\mathcal{L}$ and the upwash variance at $e = 0$ fixes the amplitude—so the frozen baseline *is* the standard Amiet–von Kármán prediction, not a recalibration of it: the $e = 0$ member of the representation is the isotropic von Kármán spectrum itself, and it reproduces the analytic one-dimensional form to within the quoted fit residual (ratio 0.79–1.14 across 100 Hz–1 kHz), with the Liepmann spectrum shown as an independent check of the unit bridge. Against this baseline the computed distortion lowers the predicted level by 5 dB at $e = 0.5$, 9 dB at $e = 1$ and 14 dB at $e = 2$ across the resolved band ($f \lesssim 1.8$ kHz at this operating point), the spanwise-coherence drain of figure 16 now carried through geometry, aerofoil response and radiation to the observer.

The chain is then confronted with measurement. Figure 18 shows four published leading-edge-noise experiments with fully documented inflow turbulence—the NACA0012 of Paterson & Amiet (1976) at 40–90 $\mathrm{m\,s^{-1}}$ (higher speeds excluded only by their Mach number), and the flat plates of Bampanis *et al.* (2022) and Narayanan *et al.* (2015)—each with four predictions computed through the identical response: the frozen von Kármán and Liepmann inflows of standard practice, and the model's computed spectra, standard and clock-relaxed, evaluated at the effective free-distortion strain accumulated along the potential-flow stagnation streamline down to the eddy scale ($e_{\mathrm{eff}} = 0.09$–0.16 for the NACA0012; below 0.05 for the thin plates). Three statements result. First, the assembled

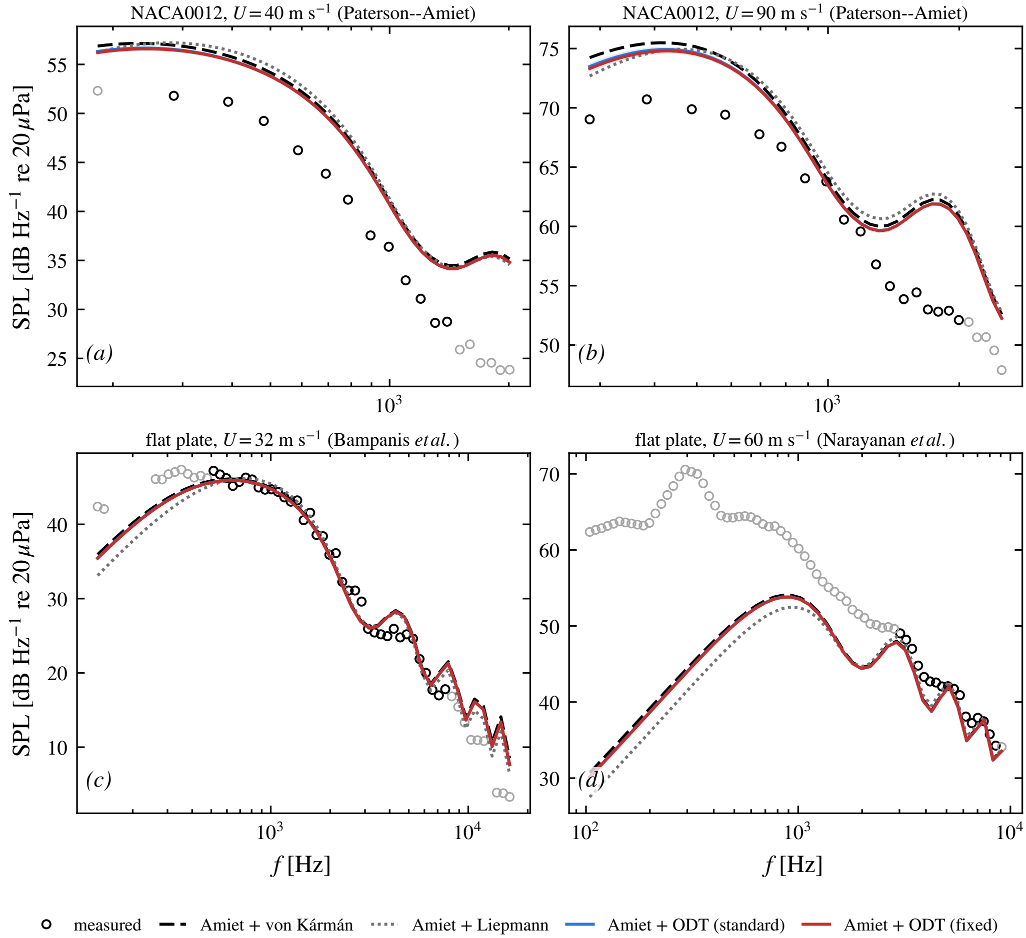

Figure 18. Absolute far-field predictions against published measurements, all four inflow spectra through the identical non-compact response (Amiet 1975; Roger & Moreau 2010): frozen von Kármán (dashed), frozen Liepmann (dotted), the computed standard-model spectrum (blue) and the computed clock-relaxed spectrum (red), each entered at the case's effective free-distortion strain. (*a,b*) NACA0012 of Paterson & Amiet (1976) ($c = 0.23$ m, $u'_w/U \approx 4$–5%, $\Lambda_f \approx 30$ mm, observer 2.25 m overhead; Gershfeld thickness admittance applied identically to all four chains); (*c*) ECL flat plate of Bampanis *et al.* (2022) ($c = 0.10$ m, 3 mm thick, $Tu = 4.5\%$, $\Lambda = 9$ mm, $\Theta = 90°$, $R = 1.25$ m); (*d*) ISVR flat plate of Narayanan *et al.* (2015) ($c = 0.15$ m, $Tu = 2.5\%$, $\Lambda = 6$ mm, $R = 1.2$ m). Black measured symbols lie in each case's stated comparison band (grey outside: background flags, facility contamination, thickness-breakdown frequency); spectra digitised from the published figures with per-figure calibration checks (± 0.5 dB). The effective strain is a geometric estimate (potential-flow stagnation-line accumulation truncated at the eddy scale); the shaded band spans truncation at $\Lambda/4$–Λ.

chain is quantitatively right where the experiment is cleanest: on the flat-plate cases every formulation sits within 1–2 dB rms of the measured spectra with the non-compactness interference lobes reproduced—the fidelity Bampanis *et al.* (2022) report for their own implementation—so the comparison discriminates inflow physics, not implementation. Second, the computed distortion behaves causally: for the thin plates the effective strain is negligible and the model correctly predicts *no* correction to the frozen spectrum, while for the NACA0012 it lowers the prediction towards the data at every speed (bias $+5.7 \rightarrow +5.3$ dB at 40 m s^{-1}, $+1.8 \rightarrow +1.4$ dB at 60, $+5.1 \rightarrow +4.7$ dB at 90), by the modest amount its geometric effective strain implies; the shift between predictions is exact, since the

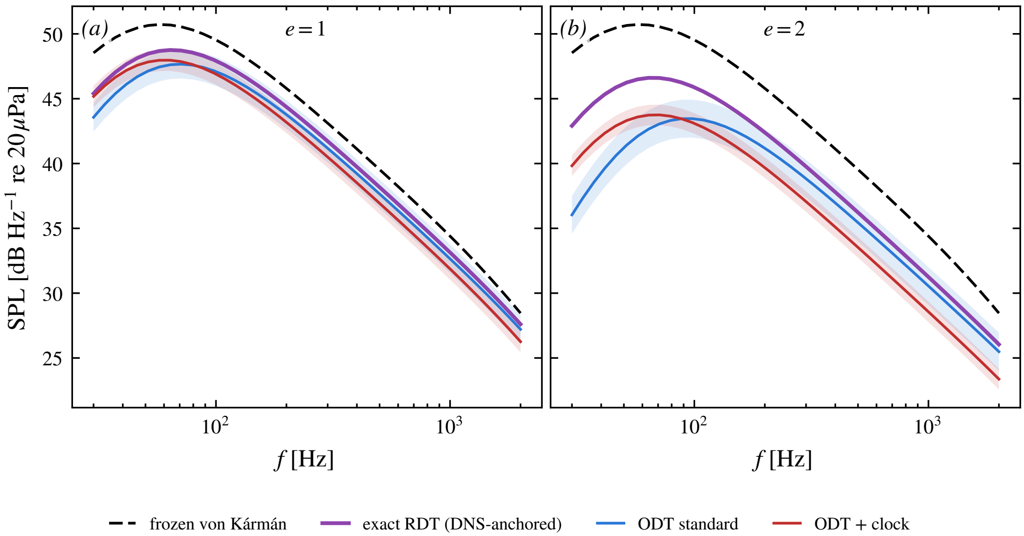

Figure 19. Computed against prescribed inflow at application strains, judged by the linear reference: absolute SPL at the reference configuration for the rapid family ($Sk_t/\varepsilon \approx 8$) at (*a*) $e = 1$ and (*b*) $e = 2$. Purple: exact (inviscid) RDT of the model's relaxed initial state—the reference the strained-box DNS of § 4.6 validates at rapid rates—with level set by the kernel's exact upwash amplification (reproducing the moment-level value to 0.7%). Dashed: the frozen von Kármán inflow of standard practice. Blue/red: the standard and clock-relaxed computed inflows (fitted representations, measured ensemble levels), each shaded by its measured spectral rms distance from the RDT family mapped to decibels.

digitisation uncertainty (±0.5 dB) is common to all four. Third, the standard and clock-relaxed models are indistinguishable in the far field at these laboratory strains (differences below 0.05 dB), and this is stated as a limit of the test, not of the mechanism: laboratory rigs are designed to minimise inflow distortion, so no published far-field measurement samples the strains at which the relaxation mechanism acts. The residual NACA0012 overprediction is the finite-thickness effect Paterson & Amiet (1976) themselves identified (breakdown near $U/ft_A \approx 1$) and Gershfeld (2004) modelled, of a piece with the blocking mechanism excluded from the present scope; it is unchanged in rank by any choice of inflow spectrum.

Where the mechanism does reach the prediction is shown in figure 19, and the claim is worded exactly as the evidence permits. At total strains of 1–2 the frozen-spectrum error against the linear reference grows from +1.3 to +3.6 dB, while both computed inflows track the reference within 1–2.5 dB—a residual that contains the physical relaxation at finite rapidity as well as the model deficiency measured in § 4.6.2—so an evolved inflow beats a prescribed one at precisely the strains where the correction matters, whichever variant of the model evolves it. Between the two variants the radiated *level* does not discriminate: the clock acts below its size cap, where the noise kernel carries little weight, and its gain appears instead in the uncertainty the input carries into the prediction—the clock-relaxed ensembles lie closer to the exact-RDT-distorted family by a factor of two to four in rms at both rapidities, and, unlike the standard model's, their distance does not grow with strain (supplementary material, § S3), which halves the shaded band of figure 19 and makes it strain-independent. A far-field discrimination of the mechanism requires a configuration whose acoustic response samples the sub-cap scales at accumulated strains of order unity—turbulence-ingesting rotors, where the streamtube contraction supplies the strain and the blade response the fine-scale weighting—and is left as the falsifiable target of the application study.

The scope of this statement is set out as carefully as its content. It concerns the *free* distortion by the mean strain: the surface-blocking effect, which measurements and simulations show *enhancing* the near-edge upwash at low frequency (Hunt & Graham 1978; dos Santos *et al.* 2023; Piccolo *et al.* 2024), is a distinct and partially compensating mechanism outside the present line model, and the strain history along individual streamlines is replaced by the canonical constant-rate distortion. What the calculation establishes is therefore not a noise prediction but the magnitude and sign of the term that frozen-spectrum practice omits: order 5–20 dB at the energy-containing reduced frequencies for total strains of 0.5–2—far too large to neglect, and of the opposite sign to the blocking correction with which it must ultimately be combined. Figure 17 shows that the chain from model output to absolute far-field level is complete and cheap; what remains for a predictive comparison against measurement is the blocking correction and the streamline strain history, for which the present model supplies the distorted-inflow input at negligible cost, with its uncertainty quantified by the envelope of § 4.6.2 and its fidelity improved by the mechanism of § 6 precisely in the wavenumbers the response weights.

8. Conclusions

We have presented a strain-coupled formulation of one-dimensional turbulence and, around it, a systematic account of what a single line of turbulence can and cannot say about rapid distortion. The formulation splits the distortion physics by structure: production enters as an exact continuous forcing, the rapid pressure–strain as an energy-conserving operator built from an explicit spectral closure, and the wavevector kinematics as a dilatation of the domain, while the eddy events of the standard model continue to carry the cascade and the slow return to isotropy. The construction preserves the model's structural invariants, reproduces the constant-free rapid-distortion onset rate identically, and is verified in the solver to the level of the closure it evolves (10^{-4} in the component fractions); at the moment level it reproduces the strained components of the Lee & Reynolds (1985) DNS quantitatively, with the one exception—the neutral-direction anisotropy—traced to the orientation information that every single-point closure discards, not to the line.

Three findings shape how such a model must be judged. First, the correct linear reference for line-projected spectra is broadband: exact rapid-distortion theory, projected onto a line, redistributes the projected upwash spectrum by a factor of 2.4 across the resolved range at $e = 1$, so a rigid spectral translation—the model's own kinematics—is not the linear prediction, and comparisons built on it conflate a structural approximation with nonlinear physics. Second, against a strained-box DNS with exact-RDT companions reduced to the same observables, the strain-coupled model carries the correct integrated anisotropy but allocates it uniformly over wavenumber, where linear theory and DNS concentrate it at the energy-containing scales; the deficiency is localised in the wavenumber-uniform rapid operator and grows with strain rapidity, a measured validity envelope that accompanies the model into any application. Third, the single-line closure question admits an exact, falsifiable answer in place of a closure claim: line data bound the rapid term sharply, the bounds are honest up to $e \approx 0.25$–0.5 under plane strain and at all computed strains under axisymmetric distortion, and the axisymmetry hypothesis is measured rather than assumed—its plane-strain violation grows with accumulated strain and rapidity while the acoustically relevant upwash amplification remains robust to it.

The measured limits point to a specific missing ingredient, which the paper identifies and supplies. The strain-induced fine-scale anisotropy is delivered directly by large-eddy maps, which is why no acceptance-gating or map-reweighting scheme can control it; what does move it is relaxation applied at the scale of the imbalance—the model's own energy-equalising kernel, applied hierarchically to eddy sub-scales. The intervention family maps the mechanism's envelope precisely: depth that reaches the measured scales is necessary and sufficient at moderate strain; dosage and delocalised application raise the early gains in proportion; and the fine-scale fade beyond $e \approx 3$, common to every scheme locked to the eddy clock, is removed by giving the relaxation its own clock—an independent stream of relaxation events whose size cap preserves the one-point physics. What remains is not timing but allocation: isotropisation is bounded at component equality and cannot reach the fine-scale sign reversal the DNS exhibits, so the remaining step is the wavenumber- and angle-dependent spectral kernel derived in § 5; exercising it in place of the wavenumber-uniform closure is the natural continuation, now motivated by a measured deficit and pointed at a measured target.

For the application that motivated the work, the model delivers what the prediction chain lacks: a distorted, anisotropic, scale-resolved upwash spectrum, evolved rather than prescribed, at two orders of magnitude below the cost of a strained DNS ensemble—and § 7 carries it through Amiet's theory to the radiated level. The assembled chain reproduces four published far-field experiments to 1–2 dB rms where the facilities are cleanest, with the computed correction causal—vanishing for thin plates, lowering the prediction towards the data for the NACA0012—and, against the linear reference at application strains, an evolved inflow beats the frozen one by a margin that grows with strain. Relative to the frozen-spectrum input of standard practice the computed distortion is a weaker leading-edge source at every resolved frequency—by 5–20 dB at the energy-containing scales for total strains 0.5–2, in the geometry-free kernel ratio and in the assembled absolute prediction at the reference configuration alike—because the distortion drains exactly the spanwise-coherent upwash content the aerofoil responds to. The omitted term is far too large to neglect and opposite in sign to the surface-blocking correction with which a complete prediction must combine it; predictions inherit a spectral-shape uncertainty of order the distance envelope at the operating point's strain and rapidity, stated here rather than discovered later.

Appendix A. The forward map and the bound kernels

This appendix derives the three ingredients that § 5 uses: the two-scalar representation of the axisymmetric solenoidal spectrum with its realisability cone, the forward map from the two scalars to the line spectra (with the falsifiers T1–T2), and the closure kernels (5.7)–(5.9), together with the linear programs that turn these linear relations into the sharp bounds of § 5.4.

A.1. *The axisymmetric representation and realisability*

The spectrum tensor of statistically homogeneous turbulence is Hermitian, positive semi-definite at each $\boldsymbol{\kappa}$, and solenoidal, $\kappa_m \Phi_{mn}(\boldsymbol{\kappa}) = 0$; it therefore has rank at most two, supported on the plane normal to $\boldsymbol{\kappa}$. Under A1 (axisymmetry about $\boldsymbol{e}_2$, mirror-symmetric, helicity-free) it is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950; Lindborg 1995):

$$\Phi_{mn}(\boldsymbol{\kappa}) = a(\kappa_2, \kappa_\perp)\, P_{mn}(\boldsymbol{\kappa}) + c(\kappa_2, \kappa_\perp)\, \xi_m \xi_n, \qquad \boldsymbol{\xi} = \boldsymbol{e}_2 - \mu \frac{\boldsymbol{\kappa}}{\kappa}, \quad \mu = \frac{\kappa_2}{\kappa}, \tag{A 1}$$

with $P_{mn} = \delta_{mn} - \kappa_m \kappa_n / \kappa^2$ the solenoidal projector, so that $\kappa_m \xi_m = 0$ and $|\boldsymbol{\xi}|^2 = 1 - \mu^2 = 1 - x$ with $x = \kappa_2^2/\kappa^2$: solenoidality is exact by construction, which is how the third would-be scalar of a general axisymmetric tensor is eliminated. For a unit vector $\boldsymbol{v}$ normal to $\boldsymbol{\kappa}$, $\Phi_{mn} v_m v_n = a + c\,(\boldsymbol{\xi} \cdot \boldsymbol{v})^2$, so the two eigenvalues of the polarisation matrix on that plane are a (attained for $\boldsymbol{v} \perp \boldsymbol{\xi}$) and $a + (1 - x)\, c$ (for $\boldsymbol{v} \parallel \boldsymbol{\xi}$). Positive semi-definiteness is therefore the pointwise *realisability cone*

$$a(\kappa_2, \kappa_\perp) \;\geq\; 0, \qquad a(\kappa_2, \kappa_\perp) + (1 - x)\, c(\kappa_2, \kappa_\perp) \;\geq\; 0, \tag{A 2}$$

quoted in § 5.4. Isotropy is the special case $c = 0$ with a a function of κ alone.

A.2. *The forward map to the line spectra*

Writing $\boldsymbol{\kappa} = (\kappa_\perp \cos\theta,\ \kappa_2,\ \kappa_\perp \sin\theta)$, the azimuthal moments at fixed $(\kappa_2, \kappa_\perp)$ are $\langle \kappa_1^2 \rangle_\theta = \langle \kappa_3^2 \rangle_\theta = \frac{1}{2}\kappa_\perp^2$, $\langle \kappa_1^4 \rangle_\theta = \langle \kappa_3^4 \rangle_\theta = \frac{3}{8}\kappa_\perp^4$, $\langle \kappa_1^2 \kappa_3^2 \rangle_\theta = \frac{1}{8}\kappa_\perp^4$, and every odd moment vanishes. Applied to (A 1) the diagonal weights average to

$$\langle P_{11} \rangle_\theta = \langle P_{33} \rangle_\theta = \tfrac{1}{2}(1+x), \quad P_{22} = 1 - x, \quad \langle \xi_1^2 \rangle_\theta = \langle \xi_3^2 \rangle_\theta = \tfrac{1}{2}x(1-x), \quad \xi_2^2 = (1-x)^2, \tag{A 3}$$

and every mixed weight to zero. Substituting into the definition (5.3) of the line spectra, $\phi_{mn}(\kappa_2) = \int_0^\infty \langle \varPhi_{mn} \rangle_\theta \, 2\pi\kappa_\perp \, \mathrm{d}\kappa_\perp$, gives the **forward map** from the two scalars to the data a line carries:

$$\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2) = 2\pi \int_0^\infty \Big[a\, \tfrac{1+x}{2} + c\, \tfrac{x(1-x)}{2} \Big] \kappa_\perp \, \mathrm{d}\kappa_\perp, \tag{A 4}$$

$$\phi_{22}(\kappa_2) = 2\pi \int_0^\infty \Big[a\,(1-x) + c\,(1-x)^2 \Big] \kappa_\perp \, \mathrm{d}\kappa_\perp, \tag{A 5}$$

$$\phi_{mn}(\kappa_2) = 0 \qquad (m \neq n). \tag{A 6}$$

Equations (A 4)–(A 6) contain the two free falsifiers quoted in § 5.4: under A1 the two transverse line spectra coincide at every resolved wavenumber (T1), and the cross-spectra vanish (T2); both are properties the line itself reports, scale by scale, with no reference to unresolved data. The ϕ_{22} bracket is the same combination of a and c that appears in the kernel g_2 below—the conventions mesh, as they must, because ϕ_{22} and the 22-component of the rapid term weight the spectrum through the same projection geometry.

They also explain why the isotropic intuition about line data fails under A1. For isotropic turbulence ($c = 0$, $a = a(\kappa)$) the map is over-determined: one unknown function of one variable against two measured functions, with ϕ_{22} determining a by the classical Abel-type inversion and the transverse spectrum slaved to it by the derivative relation. Under A1 the bookkeeping reverses: *two unknown functions of two variables* against two measured functions of one variable. At each fixed κ_2 the data supply exactly two numbers—the two weighted $\kappa_\perp$-integrals (A 4)–(A 5)—about two unknown functions of $\kappa_\perp$; any perturbation $(\delta a, \delta c)(\kappa_\perp)$ annihilating both integrals while respecting (A 2) is invisible to the line. This infinite-dimensional null space is the quantitative content of the obstruction of § 5.2 after the axisymmetric collapse.

A.3. *Derivation of the closure kernels*

For the diagonal components the exact result (5.2) reads

$$\varPi_{nn}^{(\mathrm{r})} = 2\, A_{kl} \int_{\mathbb{R}^3} \frac{\kappa_n \kappa_k}{\kappa^2} \, \varPhi_{nl}(\boldsymbol{\kappa}) \, \mathrm{d}\boldsymbol{\kappa} \qquad (\text{no sum on } n). \tag{A 7}$$

Substituting (A 1), carrying out the azimuthal average with the moments above, and specialising to $A = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ reduces each component to a $2\pi\kappa_\perp$-weighted integral over the half-plane of a polynomial combination in x, linear in (a, c)—the kernels (5.7)–(5.9) of the main text. The upwash component illustrates the mechanics: only $k = l = 1$ and $k = l = 2$ contribute, with azimuthal averages $\langle \kappa_1 \varPhi_{21} \rangle_\theta = -\frac{1}{2}\kappa_\perp^2 \kappa_2 \,[\, a + (1-x)\, c \,]/\kappa^2$ and $\langle \varPhi_{22} \rangle_\theta = (1-x)[\, a + (1-x)\, c \,]$, so that both enter through the same bracket and

$$\Big\langle \text{integrand of (A 7)} \Big\rangle_\theta \Big|_{n=2} = 2\Big[-\tfrac{1}{2} \cdot \tfrac{x(1-x)}{2} - \tfrac{1}{2}x(1-x) \Big] \big[a + (1-x)c \big] = -\tfrac{3}{2}\, x\,(1-x) \big[a + (1-x)c \big], \tag{A 8}$$

which is $g_2(x)$. The transverse components involve in addition the fourth moments of (A 3)'s underlying table and yield g_1 and g_3; the full reduction has been verified symbolically against (A 7). Three identities check the algebra: $g_1 + g_2 + g_3 = 0$ at every x for both scalars (pointwise energy conservation); each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection degenerates; and averaging over the isotropic angular measure $\mathrm{d}\mu$ on $[0, 1]$ gives $\langle g_1 \rangle : \langle g_2 \rangle : \langle g_3 \rangle = +\frac{1}{5} : -\frac{1}{5} : 0$, reproducing $\varPi_{ij}^{(\mathrm{r})} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) as in § 5.7.

A.4. *The bounds as linear programs*

The target (5.6), the data constraints (A 4)–(A 5), and the cone (A 2) are all linear in (a, c), and everything decomposes over the resolved wavenumber: writing $\varPi_{nn}^{(\mathrm{r})} = \int_0^\infty \pi_n(\kappa_2) \, \mathrm{d}\kappa_2$ with $\pi_n = 2\pi \int_0^\infty g_n(x)[a, c] \, \kappa_\perp \, \mathrm{d}\kappa_\perp$, the extremisation over all spectra consistent with the line data is a family of *independent* infinite-dimensional linear programs, one per κ_2:

$$\pi_n^{\pm}(\kappa_2) = \max/\min_{(a,c)(\kappa_\perp)} \; 2\pi \int_0^{\infty} g_n(x)\,[a,c]\,\kappa_\perp\,\mathrm{d}\kappa_\perp \quad \text{s.t.} \quad \text{(A 4)–(A 5) hold, (A 2) pointwise,} \tag{A 9}$$

whence $\varPi_{nn}^{\pm} = \int \pi_n^{\pm}\,\mathrm{d}\kappa_2$ and the indeterminacy $\mathcal{I}_n = (\varPi_{nn}^{+} - \varPi_{nn}^{-})/2k_t S$ of § 5.4, normalised by the natural rapid-term scale so that $\mathcal{I}_n$ is comparable across strains (the isotropic value is $\varPi_{11}^{(\mathrm{r})} = \frac{2}{5}k_t$ per unit $S_{11} = \frac{1}{2}$). Discretised on a logarithmic $\kappa_\perp$ grid of order 10^2 points, (A 9) is a linear program with two equality constraints and the cone, solved per κ_2 in milliseconds; the bounds are global and sharp by LP duality—no fitting and no local minima. The side conditions of § 5.4 enter as further linear constraints: the log-Lipschitz slope cap $|\mathrm{d}\ln(a,c)/\mathrm{d}\ln\kappa_\perp| \le \lambda$ and the polarisation cap $|c| \le \gamma a$. The LP extremals concentrate on few-point supports in $\kappa_\perp$ (bang–bang spectra), which is why the raw bounds are conservative and why smoothness alone barely narrows them: a log-Lipschitz spectrum can still vary like $\kappa_\perp^{\pm\lambda}$, so the angular reweighting that drives the indeterminacy survives smoothing, while the polarisation cap attacks it directly. The dual variables of (A 9) certify, per resolved wavenumber, which linear combinations of the measured line spectra control each π_n; finite-window bias on ϕ_{mn} at small κ_2 propagates linearly and can be carried as interval data in place of the equalities with no change of machinery.

Appendix B. Numerical parameters, initial conditions and statistics

B.1. *ODT configurations*

All line ensembles share the solver settings of the standard model (Stephens & Lignell 2021): eddy-rate constant $C = 7$, viscous-penalty parameter $Z = 450$, kernel parameter $\alpha = 2/3$, adaptive mesh with $\mathrm{d}x \in [2\times10^{-4}, 2\times10^{-2}]$ on a unit domain (4000 initial cells), explicit advancement with diffusive CFL 0.5 and a strain-resolved bound of 10^{-2} on the total-strain increment per step (§ 3.3). The initial condition is the resolved spectral field of § 3.3: a band-limited isotropic superposition of 64 Fourier modes peaked at eight wavelengths per domain, which places no energy at the grid scale. Strained production ensembles use an unstrained precursor: the line evolves as standard ODT until the eddy population is statistically active and the initialisation transient has relaxed, and the strain is switched on only then (§ 4.6). The molecular viscosity is set per study: 10^{-4} for the transmission diagnostic of § 6 (a scale diagnostic, resolved with wide margin), 10^{-5} for the strain-on/off ensembles of § 4.3, and 3×10^{-6} on a 16000-cell grid at the operating point of § 4.4.

B.2. *Strained-box DNS and exact-RDT companions*

The benchmark of § 4.6 is a pseudo-spectral 128^3 simulation in a Rogallo deforming frame: a decaying isotropic precursor, then plane strain $A = S\,\mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ at $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to $e = 1$ over four independent realisations, with $\kappa_{\max}\eta \approx 0.76$ at $e = 1$—a pilot resolution, so small-scale statements are qualitative. Every realisation has an exact-RDT companion: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically. The initial Fourier fields are whitened by the *symmetric* inverse square root $C^{-1/2}$ of the sample covariance, so the initial anisotropy vanishes per realisation. This detail is load-bearing: an ordered-Cholesky whitening used initially left a bias $b_{33} = +0.0148 \pm 0.0046$, detected by the nominal-turbulence null test (strain machinery active, $A = 0$) and traced by a reversal experiment to the ordering; with the symmetric form all $|b_{ii}| \le 0.005$ and the on-line falsifier T1 exceeds its confidence band at the chance rate. The estimator pipeline of § 5.4 passes the same null battery on JHTDB isotropic turbulence (1024^3, 128 lines): T1 exceedances at the chance rate, cross-spectral coherences $\sim 10^{-2}$, the longitudinal–transverse derivative relation to 4%, and the LP bounds bracketing the exact isotropic value at every constraint level.

B.3. *Statistics policy*

Spectral band quantities of the strained line are strongly intermittent across realisations—single realisations carry 10^2–10^3 times the median band energy—so ensemble means of band ratios do not converge at any practical sample size. All ODT spectral statistics are therefore medians over realisations with 95% bootstrap confidence intervals of the median, and every intervention in § 6 is assessed by the median of *paired* same-seed contrasts, quoted significant only when the bootstrap interval excludes zero. Ensemble sizes: 1024 realisations for the transmission diagnostic and the distance envelope, 128–200 for the spectrum ensembles of § 4, and four seeds for the DNS and its companions, whose derived quantities carry jackknife errors over realisations.

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